



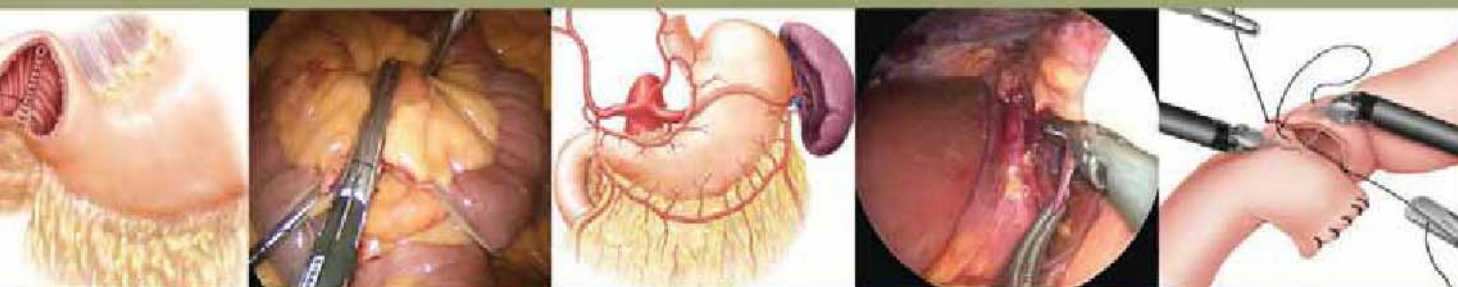
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GASTRIC SURGERY

Michael S. Nussbaum



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Master Techniques
in Surgery

GASTRIC SURGERY

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Dedication

He who combines the knowledge of physiology and surgery, in addition to the artistic side of his subject, reaches the highest ideal in medicine.

—Christian Albert Theodor Billroth

This book is dedicated to the three most important women in my life. My wife, Dr. Sue Nussbaum, inspires me daily with her wisdom and passion for always doing what is right. My wonderful and gifted daughters Jaclyn and Rachel thoroughly amaze me with their talents and creativity.

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This series of mini-atlases, of which this is the fourth, is an outgrowth of *Mastery of Surgery*. As the series editor, I have been involved with *Mastery of Surgery* since the third edition, when I joined two greats of American surgery Lloyd Nyhus and Robert Baker who were the editors at that time. At that time, in addition to *Mastery of Surgery*, which really was, almost in its entirety, an excellent atlas of how to do operations, atlases were common and some quality atlases which existed at that time by Dr. John Madden of New York, Dr. Robert Zollinger of Ohio State, and two other atlases, with which the reader may be less familiar with is a superb atlas by Professor Pietro Valdoni, Professor of Surgery at the University of Rome, who ran ten operating rooms simultaneously, and as the Italians like to point out to me, a physician to three popes. One famous surgeon said to me, what can you say about Professor Valdoni: “Professor Valdoni said to three popes, ‘take a deep breath,’ and they each took a deep breath.” This superb atlas, which is not well known, was translated by my partner, when I was on the staff at Mass General Hospital, Dr. George Nardi from Italian. Another superb atlas was that by Dr. Robert Ritchie Linton, an early vascular surgeon whose atlas was of very high quality.

However, atlases fell out of style, and in the fourth and fifth edition of *Mastery of Surgery*, we added more chapters that were “textbooky” types of chapters to increase access to the increasing knowledge base of surgery. However, atlases seem to have gone out of favor somewhat. In discussing with Brian Brown and others of Lippincott, as well as some of the editors who have taken on the responsibility of each of these mini-atlases, it seemed that we could build on our experience with *Mastery of Surgery* by having individual books which were atlases of 400 to 450 pages of high quality, each featuring a particular anatomical part of what was surgery and put together an atlas of operations of a sharply circumscribed area. This we have accomplished and all of us are highly indebted to a group of high-quality editors who will have created superb mini-atlases in these sharply circumscribed areas.

Why the return of the atlas? Is it possible that the knowledge base is somewhat more extensive with more variations on the various types of procedures, that as we learn more about the biochemistry, physiology, genetics, and pathophysiology in these different areas, there have gotten to be a variation on the types of procedures that we do on patients in these areas. This increase in knowledge base has occurred simultaneously when the amount of time available for training physicians—and especially surgeons—has been diminished and continues to do so. While I understand the hypothesis that brought the 80-hour work week upon us, and that limits the time that we have for instruction, and I believe that it is well intentioned, but I still ask the question: Is the patient better served by a somewhat fatigued resident who has been at the operation, and knows what the surgeon and what he or she is worried about, or a comparatively fresh resident who has never seen the patient before?

I do not know, but I tend to come down on the side that familiarity with the patient is perhaps more important. And what about the errors of hand off, which seem to be more of an intrinsic issue with the hand off which we are not able to really remedy entirely rather than poor intentions.

This series of mini-atlases is an attempt to help fill the voids of inadequate time for training. We are indebted to the individual editors who have taken on this responsibility and to the authors who have volunteered to share their knowledge and experience in putting together what we hope will be a superb series. Inspired by their

experience of teaching residents and medical students, a high calling, matched only by their devotion and superb care they have given to thousands of patients.

It is an honor to serve as the series editor for this outstanding group of mini-atlases, which we hope will convey the experiences of an excellent group of editors and authors to the benefit of students, residents, and their future patients in an era in which time for education seems to be increasingly limited.

Putting a book together, especially a series of books, is not easy, and I wish to acknowledge the production staff at Lippincott, Wolters Kluwer's including Brian Brown, Julia Seto, Brendan Huffinan, and many others, and my personal staff in the office who include Edie Burbank-Schmitt, Ingrid Johnson, Abigail Smith, and Jere Cooper. None of this would have been possible without them.

Josef E. Fischer, MD, FACS
Boston, Massachusetts

It is a most gratifying sign of the rapid progress of our time that our best text-books become antiquated so quickly.

—*Christian Albert Theodor Billroth*

When I was an intern at the University of Cincinnati, **Master Techniques in Surgery** series editor Dr. Josef Fischer was the chairman of the department. Every Monday afternoon we had Professor's Hour with Dr. Fischer where we would present cases to him and he would use the opportunity to teach us the nuances of surgical care that is exemplified in this series. One day, early in that year, one of the chief residents presented a case of giant duodenal ulcer; an entity that was new to me. Near the end of the session, Dr. Fischer asked for a resident volunteer to review the Cincinnati experience with giant duodenal ulcer. Fellow resident Mark Schusterman and I volunteered and thus began my academic career as a gastrointestinal surgeon.¹ Flash forward 30 years and I am now a department chair, holding my own Professor's Hour on Monday afternoons. Once again I find myself completing a project for Dr. Fischer related to gastric surgery as both editor and author in **Master Techniques in Surgery: Gastric Surgery** (see Chapter 14. *Operation for Giant Duodenal Ulcer*).

I trained during the era when gastric operations were common. Peptic ulcer and the inherent complications were frequent problems that we faced on a daily basis. The training opportunities for residents in gastric surgery have diminished over the ensuing years due to the advent of antisecretory therapy, the discovery of *helicobacter pylori*, and advances in therapeutic endoscopy. However, as this volume exemplifies, benign and malignant gastroduodenal disease continues to provide challenges to the practicing surgeon. A thorough knowledge of the various approaches and techniques are essential components in anyone's armamentarium. Further confounding these issues are the growing opportunities to apply minimally invasive approaches in certain circumstances. To be a complete gastric surgeon, one must be comfortable with both minimally invasive and open approaches, depending upon the underlying situation. In this text, where appropriate, we detail both the open and minimally invasive approaches, including laparoscopic as well as robotic applications.

The objective of the **Master Techniques in Surgery** series is to provide authoritative descriptions and illustrations of the management of each theme. This volume serves to supplement and augment the other resources available to surgeons-in-training as well as practicing surgeons faced with gastroduodenal pathology. The authors represent a truly global expertise in this field and they address the many operations and approaches in the management of benign, functional, and malignant gastric pathology. **Master Techniques in Surgery: Gastric Surgery** is intended to provide detail and clarity in the approaches to very specific diagnoses and procedures. Each chapter makes liberal use of color illustrations and photographs to elucidate important anatomical relationships and key operative steps. Further, the book comes with an associated Web site with fully searchable text and procedural videos. This volume should be useful for comprehensive review as well as a ready preoperative resource in preparation for specific gastric operations.

¹Nussbaum, M.S., Schusterman, M.A.: Management of giant duodenal ulcer. *Am. J. Surg.* 1985;199:357–361.

Once again, I am indebted to my mentor and colleague, Josef Fischer, for asking me to take on this task and allowing me to contribute as the editor of this important volume in the **Master Techniques in Surgery** series. Under Dr. Fischer's stewardship, **Mastery of Surgery** continues to withstand the test of time as an essential resource for any serious student of surgery. The **Master Techniques in Surgery** series will certainly add to that predominance as both an enhancement as well as an amplification of the body of knowledge on the subject matter.

Finally I want to thank Brian Brown and Editorial Product Manager Brendan Huffman at Lippincott, Williams & Wilkins as well as Associate Project Manager Ruchira Gupta at Aptara, Inc. for all of their insight, diligence, and support of this project. Their help and advice was invaluable to all of the authors and me. This has been a true labor of love and I hope that you find the book to be enlightening, informative, and enjoyable.

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May 2012

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1 Truncal Vagotomy with Gastrojejunostomy

Richard H. Turnage



INDICATIONS/CONTRAINDICATIONS

Indications

Gastrojejunostomy is most commonly used to treat patients with gastric outlet obstruction (GOO) of whom nearly two-thirds will have malignancies of the distal stomach, pancreas, or duodenum. The most common benign causes of GOO are peptic ulcer disease (PUD), caustic strictures, and Crohn's disease. Less common benign etiologies include tuberculosis, chronic pancreatitis, benign gastric polyps, and Bouveret's syndrome (i.e., obstruction of the pylorus by a gallstone). Although not amenable to gastrojejunostomy, systemic medical diseases, most notably diabetes mellitus, are important causes of impaired gastric emptying.

Complete resection is the treatment of choice for patients with upper gastrointestinal (UGI) malignancies. Unfortunately, locally advanced or metastatic disease precludes this approach in a significant percentage of these patients. Although noncurative gastric resection is an important option for managing symptomatic patients with advanced gastric cancer, this approach is less often feasible for patients with advanced pancreatic cancer. Endoscopic stenting of malignant obstructions of the distal stomach and duodenum is an important option in managing patients with GOO who have relatively short life expectancies, such as those with metastatic disease or poor performance scores. Advances in imaging with multidetector computed tomography have significantly improved preoperative identification of patients with advanced disease such that gastrojejunostomy is now used most often for symptomatic patients with unresectable cancer for whom endoscopic stenting has been unsuccessful or patients who are unexpectedly found to have advanced disease during exploratory laparotomy for cure.

PUD is the most common benign cause of mechanical GOO. Medical management combined with endoscopic balloon dilation results in durable relief of obstructive symptoms in about 70% of cases. This approach is particularly effective for patients with GOO due to *Helicobacter pylori*-associated ulcers. In contrast, NSAID-induced ulcers seldom resolve with nonoperative approaches. Young age, long duration of

symptoms, the need for repeated endoscopic balloon dilatations, and the continuous use of NSAIDs predict the need for surgical management. Surgical approaches to patients with GOO from PUD include truncal vagotomy and antrectomy or truncal vagotomy and a drainage procedure (either pyloroplasty or gastrojejunostomy). Truncal vagotomy and antrectomy has the lowest rate of recurrent ulcer disease. The magnitude of the operation and the consequent operative risk necessitates other strategies in patients with significant medical comorbidities as well as those patients who are found intraoperatively to have significant inflammation and scarring of the distal stomach and duodenum. Truncal vagotomy and pyloroplasty is suitable for patients with minimal inflammation or deformity of the pylorus or proximal duodenum, thereby permitting a tension-free transverse closure of the pylorus. Unfortunately, this is often not the case, making truncal vagotomy and gastrojejunostomy (TV & GJ) the best option.

Lastly, the ingestion of strong acidic solutions causes strictures in the prepyloric region of the stomach leading to GOO. Kochhar et al. have described excellent short- and intermediate-term outcomes with endoscopic balloon dilation of caustic-induced gastric strictures. Others have managed these patients with distal gastrectomy. Thus, gastrojejunostomy is reserved for those patients with prepyloric strictures who have failed endoscopic balloon dilation and are poor operative candidates for distal gastrectomy.

Contraindications

The principal contraindications to the performance of a TV & GJ are technical. If a truncal vagotomy is required in a patient who has had a previous operation in the esophageal hiatus (e.g., fundoplication or prior vagotomy), a thoracoscopic approach to vagotomy may be preferable to allow dissection in tissue planes free from scarring and inflammation. In patients with ulcer disease or other benign causes of GOO, the enlarged stomach makes the performance of a tension-free gastrojejunostomy relatively straightforward. However, peritoneal metastases in patients with gastric or periampullary cancers may cause foreshortening of the small bowel mesentery, rendering it difficult to achieve a tension-free anastomosis between the stomach and the jejunum.



PREOPERATIVE PLANNING

The initial management of a patient with GOO consists of intravascular volume and electrolyte resuscitation, nutritional repletion, and identification of the underlying cause of the obstruction. Significant intravascular volume depletion and electrolyte abnormalities, especially hyponatremia, hypokalemia, and hypochloremia, are common and are readily corrected by the intravenous infusion of a normal saline solution with potassium chloride. The loss of gastric secretions from persistent vomiting may also cause a significant metabolic alkalosis which will correct with the administration of normal saline and potassium chloride. Gastric aspiration via a nasogastric (NG) tube will decompress the dilated atonic stomach and improve the patient's abdominal discomfort, nausea, and vomiting. The intravenous administration of a proton pump inhibitor will reduce gastric acid secretion, and the administration of total parenteral nutrition via a central venous catheter will begin to correct the patient's calorie and protein deficits.

After resuscitating the patient, esophagogastroduodenostomy should be performed to determine the etiology of the obstruction. An ulcer or mass in the stomach or a suspicious lesion in the duodenum necessitates endoscopic biopsy. Biopsy of the gastric mucosa will document the presence of *Helicobacter pylori*—eradication of which will facilitate healing of benign ulcers and in many cases relieve the patient's obstruction. If a malignancy is identified or if endoscopy is indeterminate, computed tomography will characterize the lesion by determining its size, relationship to surrounding structures, and the presence or absence of visceral metastases.



SURGERY

Patients should receive a prophylactic dose of antibiotics intravenously 30 to 60 minutes prior to the incision. It is the author's preference to use cefoxitin in those patients who are not beta lactam allergic. Subcutaneous injection of 5,000 units of heparin sulfate administered preoperatively will reduce the risk of postoperative pulmonary embolism. The risk of aspiration during the induction of anesthesia and intubation may be diminished by the evacuation of gastric contents via the NG tube and posterior pressure on the cricoid cartilage.

Positioning

TV & GJ may be performed by either a laparoscopic or an open approach. Depending on the surgical approach, the patient can be placed in the supine or lithotomy position. For minimally invasive gastric approaches, if the patient is in a supine position, the operating surgeon stands on the right side of the patient while the assistant is on the contralateral side. If the lithotomy position is used, the surgeon stands between the legs. The patient is secured to the table with two safety straps, a foot board, and all of their bony prominences are well padded. For an open approach the patient is placed in the supine position with the head of the operating table elevated to facilitate exposure of the organs within the upper abdomen. The lower chest and entire abdomen is prepped with a chlorhexidine-alcohol solution and then draped with sterile towels and sheets.

Technique

The open approach will be presented in detail, but, ultimately, the same operative steps are generally true for a laparoscopic approach (for details on placement of laparoscopic trocars and specific techniques see Chapters 4, 6, and 7). When the laparoscopic approach to this procedure is used, a low threshold for conversion to an open procedure should exist when visualization of important structures are impaired. Standing on the patient's right side, the surgeon makes an upper midline incision from the xiphoid to the umbilicus. The incision is extended through the subcutaneous tissue and linea alba until the peritoneal cavity is entered. The ligamentum teres is ligated with 2-0 silk and divided and the falciform ligament is released from the anterior abdominal wall. The abdominal cavity is carefully explored and then attention is directed to the upper abdomen. In patients with GOO from ulcer disease, the location of the ulcer, extent of inflammation, and the patient's general medical condition will influence the choice of gastric resection versus a drainage procedure. For patients with malignancies, the first steps of the operation are to determine the resectability of the tumor and document the presence or absence of peritoneal or visceral metastases. The ability to mobilize an adequate length of jejunum for the performance of a tension-free gastrojejunostomy is ascertained at this time.

Truncal Vagotomy

A self-retaining abdominal wall retractor is placed to allow lateral retraction of the rectus abdominis muscles and anterior and superior retraction of the costal margins. The left lobe of the liver is mobilized by incising the avascular left triangular ligament with an electrocautery. This is facilitated by downward traction on the left lobe of the liver with the right hand while the index finger is placed behind the triangular ligament to protect the underlying structures. The triangular ligament is incised from lateral to medial until the left lobe of the liver can be folded upon itself thereby exposing the underlying gastroesophageal junction. The liver is covered with a moist laparotomy sponge and a Harrington-type blade for the self-retaining abdominal wall retractor is used to hold the left lobe of the liver to the patient's right side. In a laparoscopic approach the triangular ligament is left intact and the lateral segment of the left lobe of the liver is elevated with a self-retaining retractor such as a Nathanson retractor.

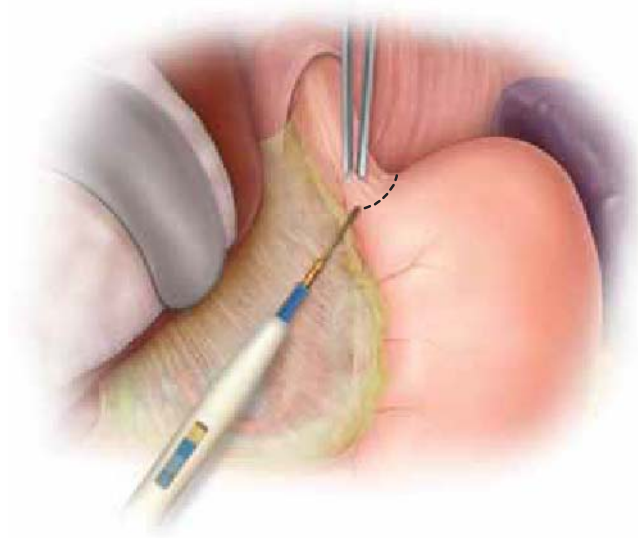


Figure 1.1 Transverse incision of the peritoneum and phrenoesophageal ligament to expose the underlying intra-abdominal esophagus and esophageal hiatus.

The peritoneum overlying the distal esophagus is grasped with DeBakey forceps and incised perpendicular to the long axis of the esophagus with Metzenbaum scissors or the electrocautery. The peritoneum and the underlying phrenoesophageal ligament are incised sufficiently to allow identification and mobilization of the esophagus (Fig. 1.1). This will usually include opening the medial most aspect of the gastrohepatic ligament on the right. The left gastric artery and its esophageal branch are left undisturbed. It is not necessary to divide the short gastric vessels on the left. By opening the incised peritoneum and phrenoesophageal ligament, the surgeon is able to free the distal 5 or 6 cm of the esophagus from the surrounding tissues by blunt dissection with his/her index finger. Once completed, the esophagus may be encircled with the index finger passed from the patient's left side to the right (Fig. 1.2). The esophagus with the NG tube is felt anterior to the surgeon's finger and the abdominal aorta is appreciated posterior. A 1-inch Penrose drain is then passed behind the esophagus to encircle it. Downward and slight anterior traction on the esophagus using the Penrose drain will facilitate blunt dissection of the distal 8 to 10 cm of esophagus from surrounding soft tissue, thus exposing the right crus of the diaphragm and the tissue lateral and posterior to the esophagus.

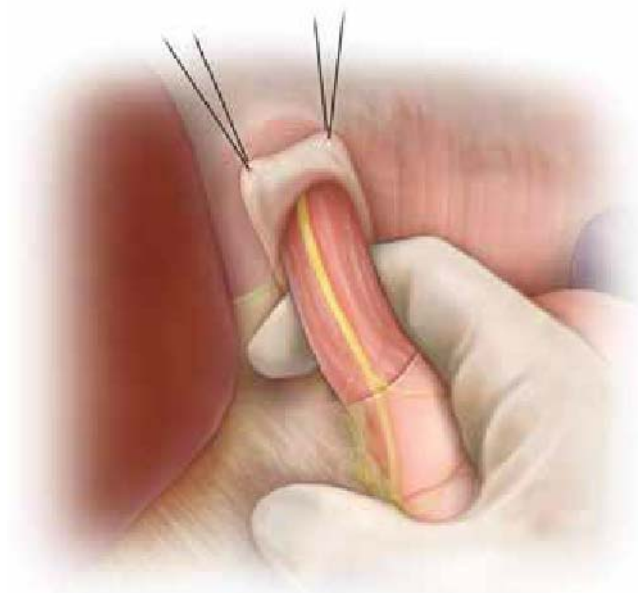


Figure 1.2 Mobilization of the distal 5 to 6 cm of esophagus from the soft tissue of the esophageal hiatus by passing the surgeon's index finger from the left to the right side of the esophagus.

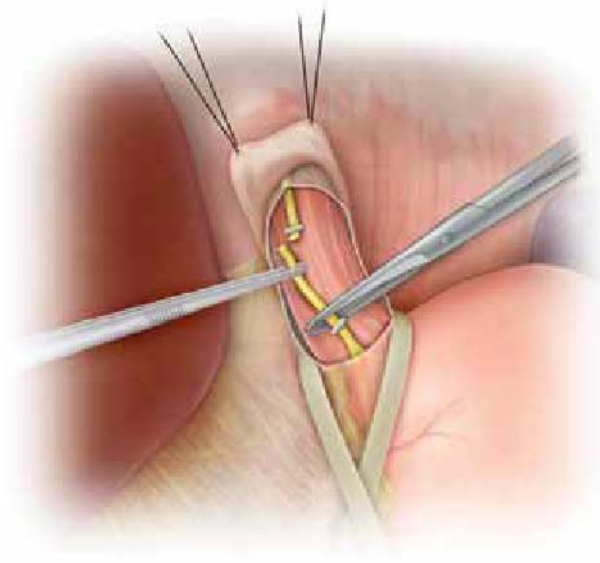


Figure 1.3 Identification of the left vagus nerve on the anterior surface of the esophagus, just to the right of the midline. The nerve is clipped with a medium stainless steel clip and then divided.

Identification of the vagus nerves is accomplished by passing the tip of the index finger over the anterior surface of the esophagus while placing downward traction on the Penrose drain. The vagus nerve fibers will present as tense, wire-like structures passing parallel to the long axis of the esophagus; the nerve fibers have been likened to “banjo strings.” In addition to the larger left and right vagus nerves, there are often multiple smaller nerve fibers traversing the surface of the esophagus and the esophageal hiatus. In an anatomic study of the vagus nerve at the esophageal hiatus, Skandalakis et al. found four or more vagal trunks in 12% of their dissections.

The left vagus nerve is usually found on the anterior surface of the esophagus just to the right of the midline. A left truncal vagotomy is performed by lifting the nerve off the surface of the esophagus with a nerve hook or a long right angle clamp. It is freed of surrounding soft tissue for a distance of 5 to 6 cm with long Metzenbaum scissors. The proximal-most portion of the nerve is then clipped with a medium stainless steel surgical clip and divided at the upper margin of the dissection just distal to the clip. The distal portion of the nerve is grasped and then clipped and divided 2 to 3 cm from the cut proximal margin (Fig. 1.3).

The right vagus nerve can be found by lifting the esophagus anteriorly while maintaining downward traction on the encircling Penrose drain. Although the nerve may be found on the posterior surface of the esophagus, it is more commonly displaced posteriorly during the process of mobilizing the distal esophagus. In this case, the nerve is found to the right of the midline resting against the posterior wall of the esophageal hiatus, anterior to the aorta. The right vagus nerve is resected as described for the left vagus by elevating the nerve trunk, applying a surgical clip to the proximal-most portion of the nerve, and then excising a 2 to 3 cm section of the nerve (Fig. 1.4). The right and left vagal trunks are sent for histologic confirmation.

With the right and left vagal nerve trunks divided, a careful search for additional nerve fibers is undertaken. This is best accomplished by downward traction on the Penrose drain and careful palpation of the anterior and posterior surfaces of the esophagus. The distal 5 to 6 cm of the esophagus and the surrounding tissue must be carefully inspected to identify and divide any small nerve fibers that may have branched from the vagal trunks in the chest, such as “the criminal nerve of Grassi” which arises from the right vagal trunk to innervate a portion of the posterior gastric fundus. The Penrose drain is then removed and the esophagus allowed to fall back toward the esophageal hiatus. The operative field is then carefully inspected for bleeding which is readily controlled with the electrocautery. The incised peritoneum and phrenoesophageal ligament are not reapproximated.

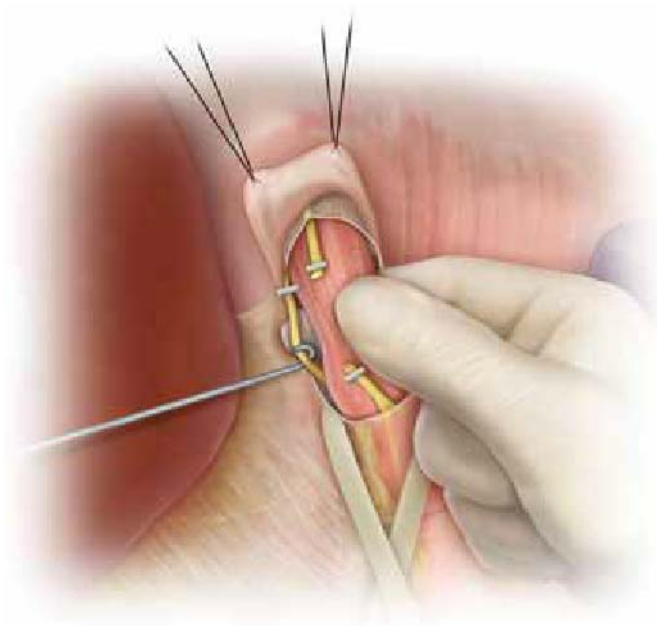


Figure 1.4 Identification of the right vagus nerve posterior to the esophagus on the right side of the esophageal hiatus.

Gastrojejunostomy

The gastrojejunostomy may be constructed on either the anterior or the posterior surface of the distal stomach. The optimal site will be influenced by the amount of scarring and deformity in the distal stomach or duodenum in patients with benign strictures and by the location of the tumor in patients with malignancies. There is no evidence that anterior, antecolic gastrojejunostomy is superior to posterior, retrocolic gastrojejunostomy or vice versa. The easier anterior gastrojejunostomy is most commonly performed and hence it is this anastomosis that is described.

Hand-sewn, Anterior or Antecolic Gastrojejunostomy

With the self-retaining abdominal wall retractor in place, the greater omentum and transverse colon are elevated to reveal the ligament of Treitz just to the left of the midline at the base of the transverse mesocolon. The proximal jejunum is brought onto the anterior surface of the stomach and oriented such that the afferent limb is adjacent to the body of the stomach and the efferent limb is juxtaposed to the distal stomach. The length of the afferent limb should be as short as possible to reduce the risk of afferent limb obstruction. The surgeon must be careful to avoid twisting of the bowel or placing tension on the anastomosis.

Stay sutures of 3-0 silk are placed through the seromuscular layers of the stomach and the jejunum at the proximal and distal ends of the anastomosis, about 8 cm apart from one another. The ends of these sutures are secured with a hemostat and brought out the left and right side of the operative field. This maneuver orients the anastomosis horizontally allowing the surgeon to sew toward himself or herself.

The back row of the anastomosis is constructed by placing interrupted 3-0 silk Lembert sutures through the serosa and muscularis of the juxtaposed stomach and the jejunum about 8 mm behind the planned gastrotomy and enterotomy. The sutures extend from the right stay suture to the left and are placed about 8 mm apart from one another. The sutures are tied after the entire row has been placed. The middle suture is not cut at this time but left long to aid in the placement of the inner posterior row of sutures. The full thickness of the stomach is then incised with the electrocautery parallel to and 8 mm from the back row of silk sutures. Fluid within the stomach is aspirated and a matching incision is made through the adjacent jejunal wall. The gastrotomy and jejunotomy should be approximately 1 centimeter shorter than the site of the silk stay sutures (Fig. 1.5).

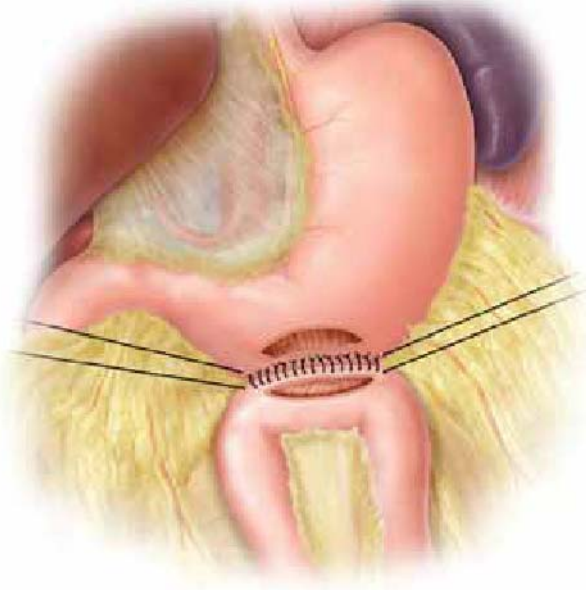


Figure 1.5 Construction of the outer layer of the back wall of an anterior or antecolic gastrojejunostomy using interrupted 3-0 silk Lembert sutures.

The inner layer of the anastomosis is constructed by sewing the juxtaposed gastric and jejunal walls together with two running, spiral sutures of 3-0 polyglycolic acid. The sutures are started at the midpoint of the back wall of the anastomosis and continued around the corners until the front wall is closed. The stitch is placed about 4 to 5 mm from the cut-edge of the bowel and incorporates the full-thickness of the gastric and jejunal walls. Elevating the serosal edge of the jejunum (or stomach) with DeBakey forceps 1 cm from the site of the planned stitch will help to incorporate all the layers of the bowel in the stitch. At the corner of the anastomosis, slightly more serosa than mucosa is included in the stitch to facilitate the invagination of the mucosa within the lumen of the anastomosis. As each stitch is placed, the edges of the anastomosis are gently drawn together. Ultimately, the inner layer of the anastomosis is completed and only the seromuscular edges of the stomach and the jejunum are visible (Fig. 1.6).

The front wall of the anastomosis is now completed by placing interrupted seromuscular Lembert sutures of 3-0 silk between the gastric and the jejunal walls. These stitches are placed several millimeters from the inner layer of the anastomosis

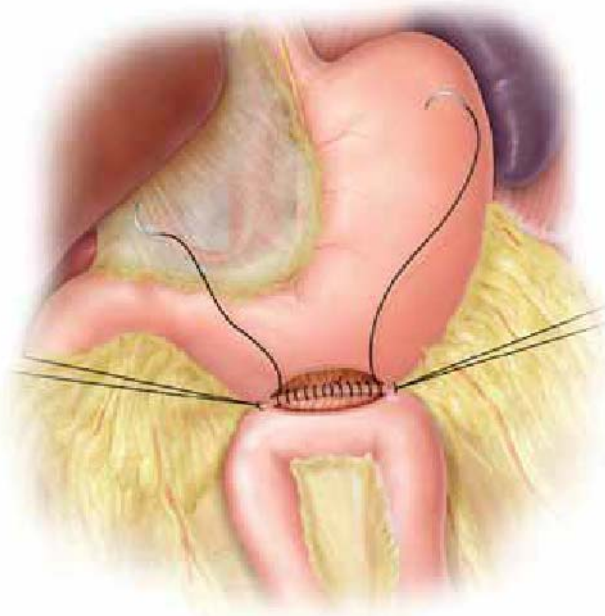


Figure 1.6 Construction of the inner layer of the back wall of anterior gastrojejunostomy using a continuous 3-0 polyglycolic acid suture placed through all layers of the juxtaposed stomach and jejunum.

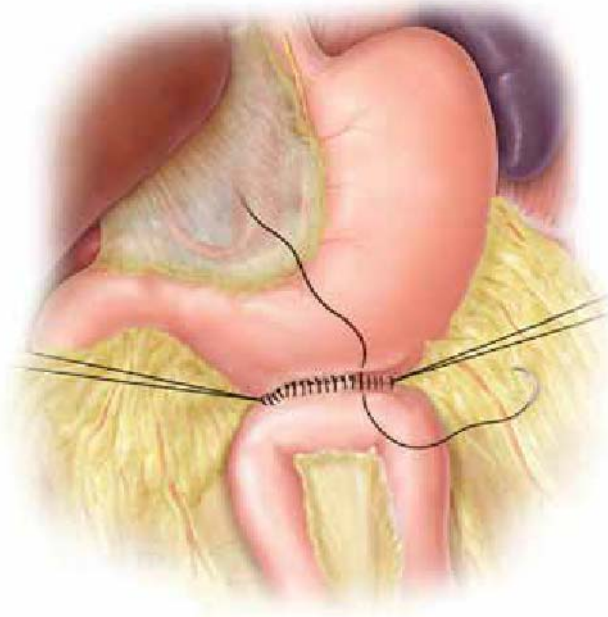


Figure 1.7 Construction of the outer layer of the front wall of the anterior gastrojejunostomy using interrupted 3-0 silk Lembert sutures placed 7 to 8 mm from one another. This row of sutures should completely bury the completed inner layer of the anastomosis.

and about 7 to 8 mm apart. This layer should bury the inner layer from view (Fig. 1.7). The corners of the anastomosis can be completed by either tying the stay sutures or by placing a new Lembert stitch in each corner to completely cover the inner layer of the anastomosis. The lumen of the anastomosis should be readily palpable either through the gastric or the jejunal wall. The operative field is inspected for evidence of bleeding or injury and the self-retaining retractor is removed. The linea alba is closed using a running suture of no. 1 polydioxanone, and the skin is closed with a skin stapler.

Stapled, Anterior Gastrojejunostomy

The orientation of the stomach and jejunum and the placement of the 3-0 silk stay sutures at the corner of the proposed anastomosis is as described earlier. A small gastrotomy and jejunotomy are made about 1 cm from the stay suture at the proximal corner of the anastomosis. One limb of a 60-mm linear stapling device (endoscopic linear stapler in laparoscopic) is inserted into the jejunum and the other limb

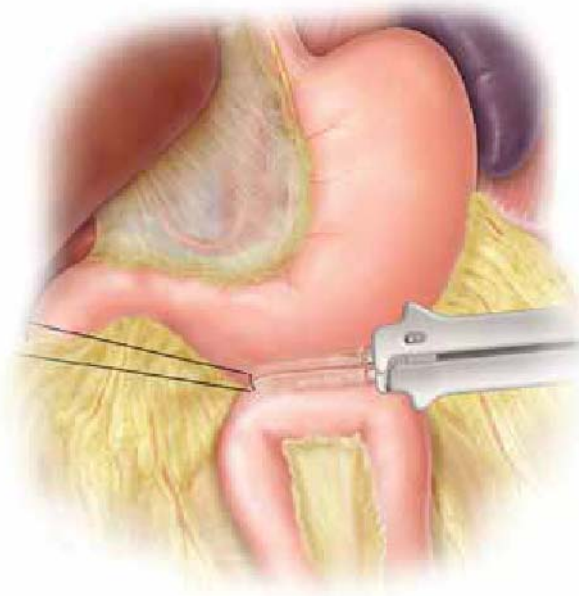


Figure 1.8 Construction of anterior gastrojejunostomy using a GIA 60 stapler inserted through a small gastrotomy and matching enterotomy. Stay stitches of 3-0 silk are placed at the proximal and distal margins of the proposed anastomosis.

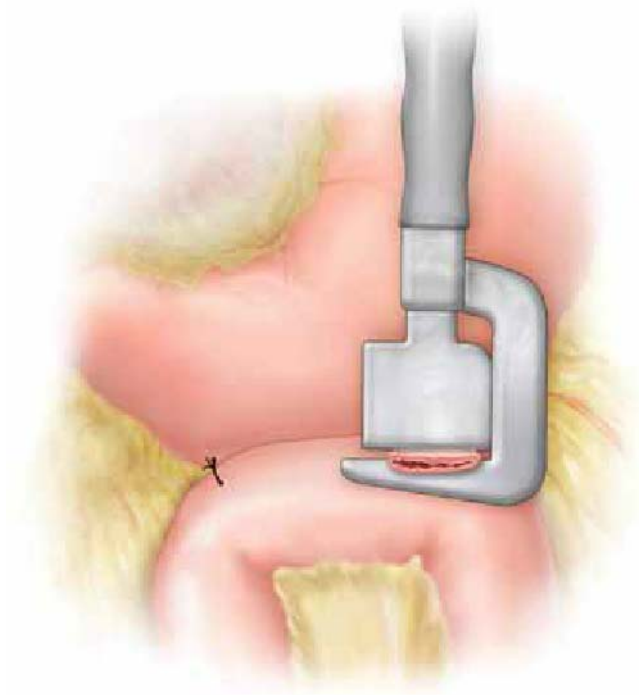


Figure 1.9 Completion of the stapled anterior gastrojejunostomy by closing the insertion site of the GIA stapler using a TA 30 stapler.

is inserted into the stomach. The device is locked and the site inspected to ensure that no tissue other than the stomach and jejunum is included within the stapler (Fig. 1.8). The stapler is then fired and removed from the operative field. The staple line is then carefully inspected for bleeding. Exposure of the staple line can be facilitated by gentle retraction with the long arm of an Army–Navy retractor. Bleeding from the staple line can be controlled with either the electrocautery or the placement of a 4-0 polyglycolic acid suture. The remaining defect in the anastomosis is grasped with several Allis clamps and a TA-30 or TA-50 stapler is aligned behind the clamps (endoscopic linear stapler or intracorporeal interrupted sutures in laparoscopic approaches). The TA stapler is then fired and the redundant tissue is excised with the scalpel before removing the stapler. The staple line is inspected for bleeding which is controlled with the electrocautery or a 4-0 polyglycolic acid suture (Fig. 1.9).



POSTOPERATIVE MANAGEMENT

Most patients undergoing a TV & GJ may be safely managed in the general surgical ward. Intravenous fluid administration with an isotonic solution for the first 24 hours will maintain a normal intravascular volume. Generally patients will mobilize their third-spaced fluids on postoperative day 3 and the infusion of 5% dextrose/0.45% sodium chloride solution will allow the maintenance of normal intravascular volume until the patient is able to resume oral intake. Antibiotics are not administered postoperatively unless there is clinical evidence of infection. Urine output may be a useful guide to the patient's volume status in the first day or two postoperatively; thereafter, the catheter is removed. The NG tube is removed when the output is less than 300 mL per 8-hour period. If the patient is nauseated or has a distended abdomen the tube is left in place for a longer period of time. As described subsequently, poor gastric emptying is a relatively common early complication necessitating longer periods of NG aspiration. A clear liquid diet is started when the NG tube has been removed and the patient is passing flatus. If liquids are tolerated for 24 hours, the patient is allowed a regular diet and the parenteral nutrition is stopped.

Adequate pain management is an important part of the postoperative care of these patients. The upper abdominal location of the incision is painful especially when the patient takes deep breaths. Poor pain relief will diminish inspiratory volume and promote the development of atelectasis and pneumonia. Patient-controlled analgesia (PCA) with morphine, hydromorphone, or meperidine provides excellent pain relief. The epigastric location of the incision makes the use of postoperative epidural analgesia less useful.

The patient is instructed in the use of an incentive spirometer preoperatively, and it is used throughout the patient's postoperative course to reduce atelectasis and minimize the risk of pneumonia. Subcutaneous heparin injections, also started preoperatively, are continued in the postoperative period to minimize the risk of pulmonary emboli. The patient is ambulated on the first postoperative day and daily thereafter. If a truncal vagotomy has not been performed, the patient should receive a proton pump inhibitor intravenously in the early postoperative period and then orally as he or she tolerates a diet.



COMPLICATIONS

The incidence of intraoperative complications during TV & GJ is low with the most common being hemorrhage from a retraction injury to the liver or spleen during the performance of the vagotomy or injury to the middle colic vein during the gastrojejunostomy. Bleeding from the liver may be avoided by careful mobilization of the left lobe and gentle retraction. Bleeding from the spleen necessitating splenectomy can be minimized by avoiding traction on the short gastric vessels or spleen.

The early postoperative complications of gastrojejunostomy for treating patients with benign or malignant GOO are shown in Table 1.1. With the exception of the study by Kim et al., most of the patients in these studies underwent gastrojejunostomy alone, without truncal vagotomy. Overall, the rate of complications following gastrojejunostomy is relatively high, ranging from 22.7 to 62.5% (median = 32%). Most complications were relatively minor, although wound infections, delayed gastric emptying, and

TABLE 1.1 Complications of Gastrojejunostomy for Gastric Outlet Obstruction in Patients with Benign or Malignant Diseases

Study Design	Sample Size	Mortality Rate (%)	Morbidity Rate	Complications
Retrospective comparison of Lap TV + GJ vs. Open TV + GJ in patients with PUD with GOO, patients accrued between 1999 and 2005	8	0	62.5	Delayed gastric emptying (50%); atelectasis (12.5%)
Retrospective review of patients with Crohn's disease and duodenal obstruction, patients accrued between 1995 and 2006	11	0	36	GJ leak (9%) GJ bleeding (9%) Early SBO (9%)
Retrospective review of GJ in patients with pancreatic cancer with GOO (66 patients) and without GOO (447 patients), patients accrued between 1996 and 2010	513	1.6	36.7	Wound infections (4.3%) Pleural effusion (2.5%) Cholangitis (2%)
Prospective comparison HJ alone in patients with pancreatic cancer with GOO (HJ + GJ) and without GOO (HJ alone), patients accrued between 1992 and 1995	44	9	22.7	Delayed gastric emptying (11%) Wound infection (9%) Intra-abdominal abscess (4.5%)
Single-center randomized trial, GJ vs. no GJ in patients with unresectable pancreatic cancer without GOO, patients accrued between 1994 and 1998	44	0	32	Cholangitis (9%) Wound infection (5%) Delayed gastric emptying (2%)
Multicenter, randomized trial of GJ vs. stent for patients with GOO from UGI cancers, patients accrued between 2006 and 2008	18	0	33	Delayed gastric emptying (11%) Wound infection (5%) UTI (5%)
Prospective, randomized trial of HP + GJ vs. HJ alone for patients with advanced periampullary cancers without evidence of GOO at presentation, patients accrued between 1998 and 2002	36	3	31	Delayed gastric emptying (17%) Cardiac complications (11%) Wound infection (8%)

Lap TV + GJ, laparoscopic truncal vagotomy and gastrojejunostomy; Open TV + GJ, truncal vagotomy and gastrojejunostomy performed through a laparotomy; GOO, gastric outlet obstruction; PUD, peptic ulcer disease; GJ, gastrojejunostomy; HJ, hepatojejunostomy.

anastomotic leak or intra-abdominal abscess occur with significant frequencies. The 30-day mortality rate associated with this operation ranged from 0 to 9% (median = 0).

Long-term complications specific to TV & GJ may be categorized as physiologic or mechanical. The physiologic complications include delayed gastric emptying, early satiety and weight loss, cholelithiasis, chronic diarrhea, dumping syndrome, and iron and calcium malabsorption resulting in anemia and osteoporosis, respectively. Patients in whom vagotomy was not performed or those in whom the vagotomy was incomplete are at risk for peptic ulceration of the jejunal side of the anastomosis. This condition is termed marginal ulceration and may cause life-threatening GI hemorrhage or obstruction of the anastomosis. In patients undergoing gastrojejunostomy without vagotomy this complication is prevented by the administration of proton pump inhibitors or histamine-2 receptor antagonists. The mechanical complications of this procedure include obstruction of the afferent or efferent limb, stenosis or kinking of the anastomosis, and small bowel obstruction.



RESULTS

The results of performing a gastrojejunostomy for patients with benign or malignant GOO are shown in Table 1.2. As alluded to in the previous paragraph, delayed gastric emptying in the early postoperative period is a significant problem associated with this procedure. The median time to tolerating a diet is 8.15 days (range = 4.4 to 8.5 days). The median length of hospitalization is 12.2 days (range = 8 to 21 days). The advantage of gastrojejunostomy over the placement of endoscopic stents or endoscopic dilation is a very low rate of persistent or recurrent GOO requiring remedial therapy. The median rate of patients requiring reoperation for either persistent or recurrent GOO is 0% (range = 0 to 5% and 0 to 16.7%, respectively). Of those patients undergoing gastrojejunostomy for malignant GOO the median survival was 3.8 months (range = 2.1 to 7.2 months).

TABLE 1.2

Results of Gastrojejunostomy in Patients with Benign and Malignant Gastric Outlet Obstruction

Study Design	Sample Size	Length of Hospitalization (days)	Time to Diet (days)	Persistent GOO/Reoperation (%)	Recurrent GOO (%)	Survival
Retrospective comparison of Lap TV + GJ vs. Open TV + GJ in patients with PUD with GOO, patients accrued between 1999 and 2005	8	21	8.5	0	0	n/a
Retrospective review of patients with Crohn's disease and duodenal obstruction, patients accrued between 1995 and 2006	11	12.2	4.4	9	0	n/a
Retrospective review of GJ in patients with pancreatic cancer with GOO (66 patients) and without GOO (447 patients), patients accrued between 1996 and 2010	513	8	n/a	0	3.1	6 mo
Prospective comparison, HJ alone in patients with pancreatic cancer with GOO (HJ + GJ) and without GOO (HJ alone), patients accrued between 1992 and 1995	44	21.5	8.3	0	0	4.8 mo
Single-center randomized trial, GJ vs. no GJ in patients with unresectable pancreatic cancer without GOO, patients accrued between 1994 and 1998	44	8.5	n/a	0	0	8.3 mo
Multicenter, randomized trial of GJ vs. stent for patients with GOO from UGI cancers, patients accrued between 2006 and 2008	18	15	8	16.7	5	2.6 mo
Prospective, randomized trial of HP + GJ vs. HJ alone for patients with advanced periampullary cancers without evidence of GOO at presentation, patients accrued between 1998 and 2002	36	11	n/a	0	5.5	7.2 mo

Lap TV + GJ, laparoscopic truncal vagotomy and gastrojejunostomy; Open TV + GJ, truncal vagotomy and gastrojejunostomy performed through a laparotomy; GOO, gastric outlet obstruction; PUD, peptic ulcer disease; GJ, gastrojejunostomy; HJ, hepatojejunostomy.



CONCLUSIONS

TV & GJ is used to manage patients with GOO from PUD who have not responded to medical management and endoscopic dilation. Gastrojejunostomy (without vagotomy) is used to treat patients with UGI malignancies and impending or actual GOO who are discovered during exploratory laparotomy to have advanced or unresectable disease. Technically the operation is relatively straightforward requiring access to the esophageal hiatus and the ability to create a tension-free anastomosis between the stomach and the proximal jejunum. Early postoperative morbidity rates are relatively high when compared to the significance of these patients' underlying illness and other medical comorbidities. Delayed gastric emptying is a common complication prolonging hospital stay. Patients who undergo gastrojejunostomy without vagotomy require acid suppression to prevent marginal ulcers.

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2 Truncal Vagotomy and Pyloroplasty

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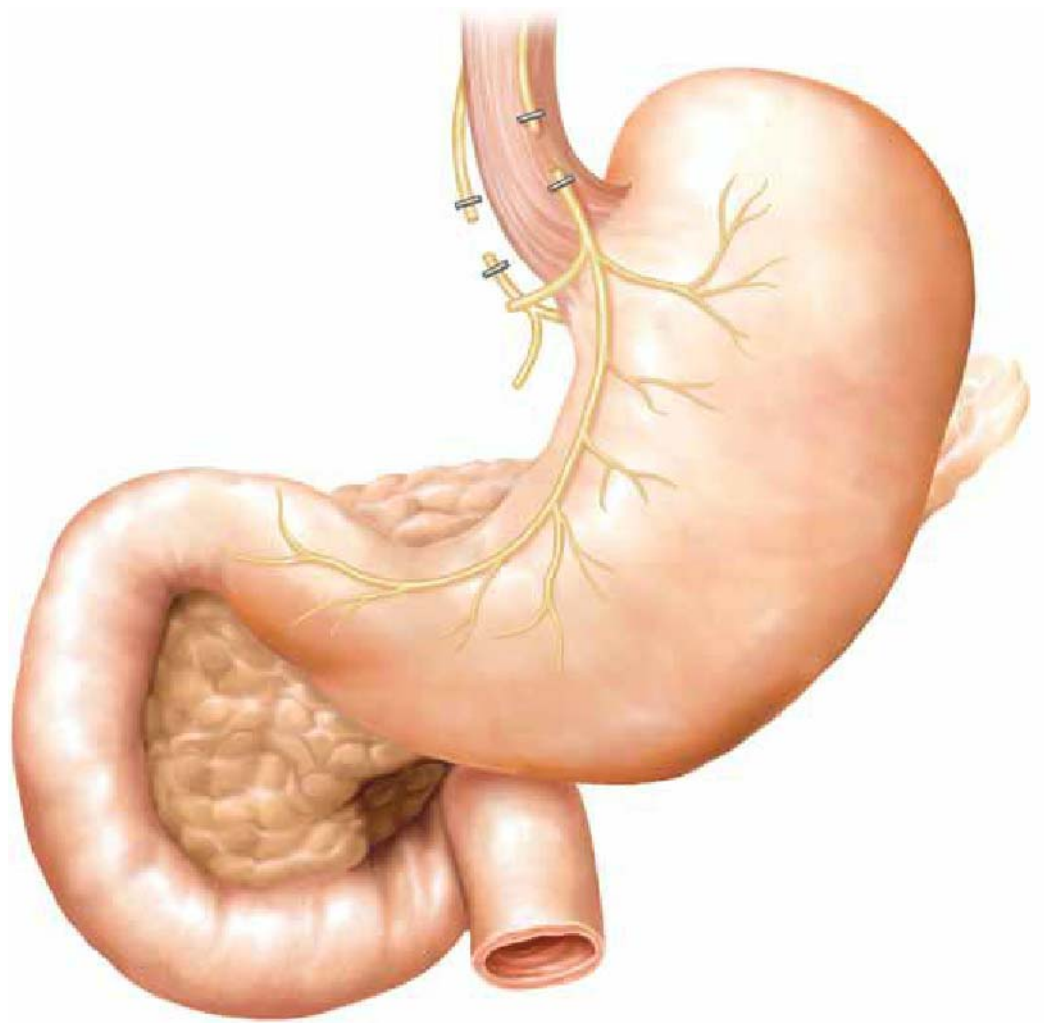
Introduction

The word *vagotomy* technically means vagal nerve transection, thus interrupting sensory and motor impulses to the stomach and gastrointestinal tract. However, a 1- to 2-cm section of each nerve is usually resected. Although the proper term for resection should be “vagectomy,” *vagotomy* is the much more commonly used term (Fig. 2.1). Its purpose is the elimination of direct cholinergic stimulation of gastric acid secretion. Vagal fibers innervate the stomach and play a major role in the cephalic phase of gastric acid secretion by their release of acetylcholine, which stimulates acid secretion from the parietal cell. The distal portion of the anterior and posterior vagal trunks also sends motor branches to the antrum and pylorus. The celiac branch of the posterior vagus stimulates small intestine motility as well. Gastric motility is affected by the antral and pyloric branches of the vagus that stimulate peristaltic activity of the antrum and pyloric relaxation. The vagus also stimulates receptive relaxation of the fundus, resulting in accommodating intake with no corresponding pressure increase.

Vagotomy results in a 75% decrease in basal acid secretion and a 50% decrease in maximum acid output. Additionally, due to the loss of reflex relaxation of the gastric fundus, increased gastric capacity after eating is attended by a rise in pressure, resulting in rapid emptying of liquids. Vagotomy also disturbs distal stomach motility, resulting in difficulty in emptying solids. Because of these changes, approximately 20% to 30% of vagotomy patients develop gastric atony, which may lead to stasis and chronic abdominal pain and distention. Hence, it is surgical dogma that after truncal vagotomy patients require a “drainage” procedure to offset the nonrelaxing, obstructing pylorus. Although four kinds of vagotomy have been described in the surgical literature—*truncal*, *selective*, *highly selective*, and *supradiaphragmatic*—truncal and highly selective vagotomy are generally used to treat peptic ulcer disease. Selective and supradiaphragmatic vagotomies are rarely used.

In Dragstedt's initial and seminal series of truncal vagotomy for the treatment of duodenal ulcer disease in 1943, nearly one-third of his patients experienced postoperative nausea, vomiting, and distention. His subsequent investigations revealed that truncal vagotomy denervated the antrum and pylorus resulting in a functional gastric

Figure 2.1 Schematic of truncal vagotomy. Both branches of the vagus nerve are ligated between clips.



outlet obstruction. It became apparent that a “drainage” procedure was necessary to avoid the symptoms of gastric stasis. Therefore, any patient who undergoes truncal, selective, or supradiaphragmatic vagotomy should undergo a drainage procedure to facilitate gastric emptying. Drainage procedures can be divided into two categories: pyloroplasties and gastrojejunostomy. Pyloroplasty is the favored approach as it maintains the original anatomy, is simple, and is accompanied by less bile reflux than gastrojejunostomy. Currently over 90% of all drainage procedures are variations of pyloroplasty. Pyloroplasty was originally described independently by two surgeons, Heineke and Mikulicz, in 1888, decades before its routine application for drainage after vagotomy. The Heineke–Mikulicz (HM) pyloroplasty is popular because it is technically uncomplicated, widely applicable, and associated with few complications. In children with pyloric stenosis, pyloromyotomy is usually performed, leaving the mucosa intact. This approach in adults is often unsuccessful as the duodenal mucosa is more adherent to the muscle layer and the intestinal lumen is often entered during the myotomy.

Management

Three different types of pyloroplasty have been described. Truncal vagotomy and pyloroplasty is a relatively simple procedure, but there are some nuances that will prevent recurrences and complications. It can be performed laparoscopically or with

open surgical technique. Indications most often include persistent ulcer disease refractory to medical therapy.

Vagotomy

Exposure is through an upper midline incision for the open technique or with five trocars positioned in a similar way as for any laparoscopic foregut operation (see Figs. 2.1 and 2.2). First, attention is turned toward the gastroesophageal junction. We use a Thompson retractor to elevate the left lateral segment of the liver when performing surgery laparoscopically, or a Bookwalter or Omni retractor for open surgery. For open surgery, the left triangular ligament is incised using cautery to allow medial retraction of the left lateral segment of the liver. This step is not necessary in the laparoscopic approach since the liver is elevated in a cephalad direction to attain access to the diaphragm hiatus. The gastrohepatic ligament is divided. Also, division of the superior attachments of the gastrosplenic ligament makes encircling the esophagus easier. The esophagus is then encircled with a Penrose drain for retraction. The crura are separated from the esophagus, and the esophagophrenic ligament is incised to expose the anterior esophagus. This step is especially important in patients with hiatal hernia. Occasionally, the gastroesophageal fat pad must be dissected if prominent, exposing the GE junction anteriorly. Again, if the lower esophagus is exposed in the mediastinum, the vagotomy may be performed superior to the GE junction fat pad.

Attention is then turned toward the anterior branch of the vagus nerve (Fig. 2.2). It is easily found traversing from left to right at the distal esophagus. Dissection of the adventitia off the anterior esophagus allows exposure of the anterior vagus nerve, which can be lifted with the help of a right-angled dissector. A short length is dissected, clips are placed proximally and distally, and a segment of the nerve is resected with scissors if clips are used. Otherwise use of ultrasonic energy facilitates ligation. If any concern exists that the resected tissue is not the vagus nerve, this small segment of tissue can be sent to pathology for frozen section to confirm nerve.

Next, the esophagus is retracted, usually anteriorly and toward the left, exposing the posterior esophagus. Since it is larger than the anterior branch, the posterior vagus nerve is usually easier to find and dissect off the esophagus (Fig. 2.3). Similar to the anterior vagus, the posterior branch is hooked with a right-angled dissector, clipped,

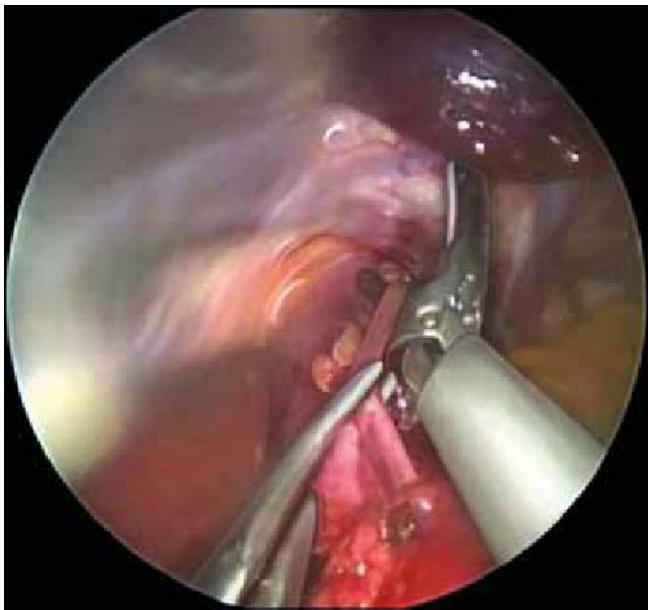


Figure 2.2 Anterior vagotomy. The esophagus is encircled with a Penrose drain for retraction. The esophageal fat pad is retracted inferiorly and the adventitia of the esophagus is cleared. The vagus nerve is dissected off the esophagus and a segment is ligated between clips or, as in this case, using ultrasonic shears.

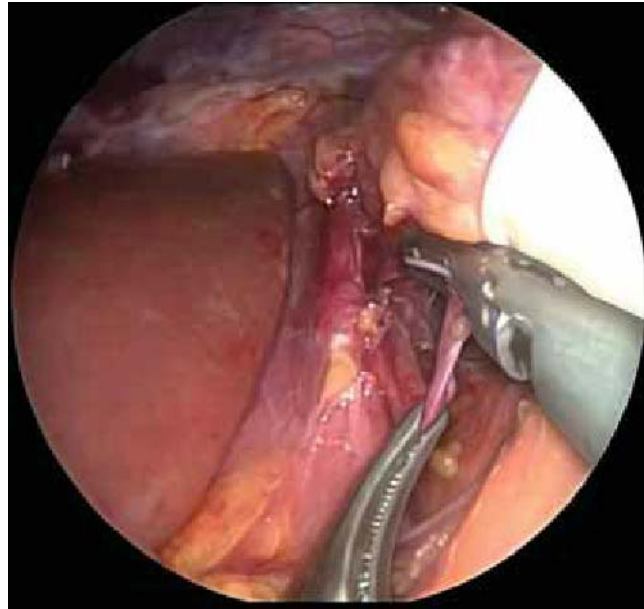


Figure 2.3 Posterior vagotomy. The esophagus is retracted anteriorly, and the vagus nerve is dissected. This image depicts the ligation of the posterior vagus nerve using ultrasonic energy.

and ligated and a portion of the nerve is resected. The Penrose drain and liver retraction are then released, and attention is turned toward the pylorus. Remember, additional vagal trunks are present in nearly 20% of patients and must be identified and transected.

Pyloroplasty

There are several methods for performing pyloroplasty. The most common is the HM pyloroplasty (Fig. 2.4). First, a Kocher maneuver is performed, mobilizing the duodenum from its retroperitoneal attachments. This ultimately helps take tension off the suture line. The pyloric muscle is easily palpated between the surgeon's thumb and index finger. Stay sutures of 3-0 silk are placed with a seromuscular bite through the pylorus just superior and inferior to the anterior aspect of the pylorus. A longitudinal incision is made in between the two stay sutures using a needle-tip electrocautery pencil or ultrasonic energy source. Care is taken to avoid making the incision longer than necessary to prevent tension on the suture line. We recommend extending it between 1 and 2 centimeters past the pyloric muscle proximally and distally (Fig. 2.5). The stay sutures are distracted, and the suture line is closed transversely (Fig. 2.6).

Many different methods exist for closing the pyloroplasty, including single-layer suture closure (with or without imbricating stitch), double-layer closure, or even stapled closure. We prefer a single-layer closure with interrupted 3-0 silk sutures using a Gambee stitch to imbricate the suture line, opposing serosa to serosa (Fig. 2.7). This allows secure closure without narrowing the lumen. Care is taken to avoid missing the mucosa, especially at the corners of the suture line as the pylorus is pushed away from the serosa by the large, thick sphincter muscle layer. If the pyloroplasty is performed laparoscopically, we prefer to use a free needle as opposed to automated suture systems for accurate needle placement. We start from the corners and work toward the middle of the suture line. If possible, the last three sutures should be tied only after all sutures are placed to facilitate assurance of mucosal bites in each stitch. The suture line can be tested by occluding the duodenum and instilling 500 to 1,000 mL of methylene blue down the orogastric tube. Omentum can be placed overlying the suture line and sutured in place if there is any concern regarding suture line integrity.

Another very quick and easy way of performing the HM pyloroplasty is with a stapling device. This is much easier to perform using open rather than laparoscopic

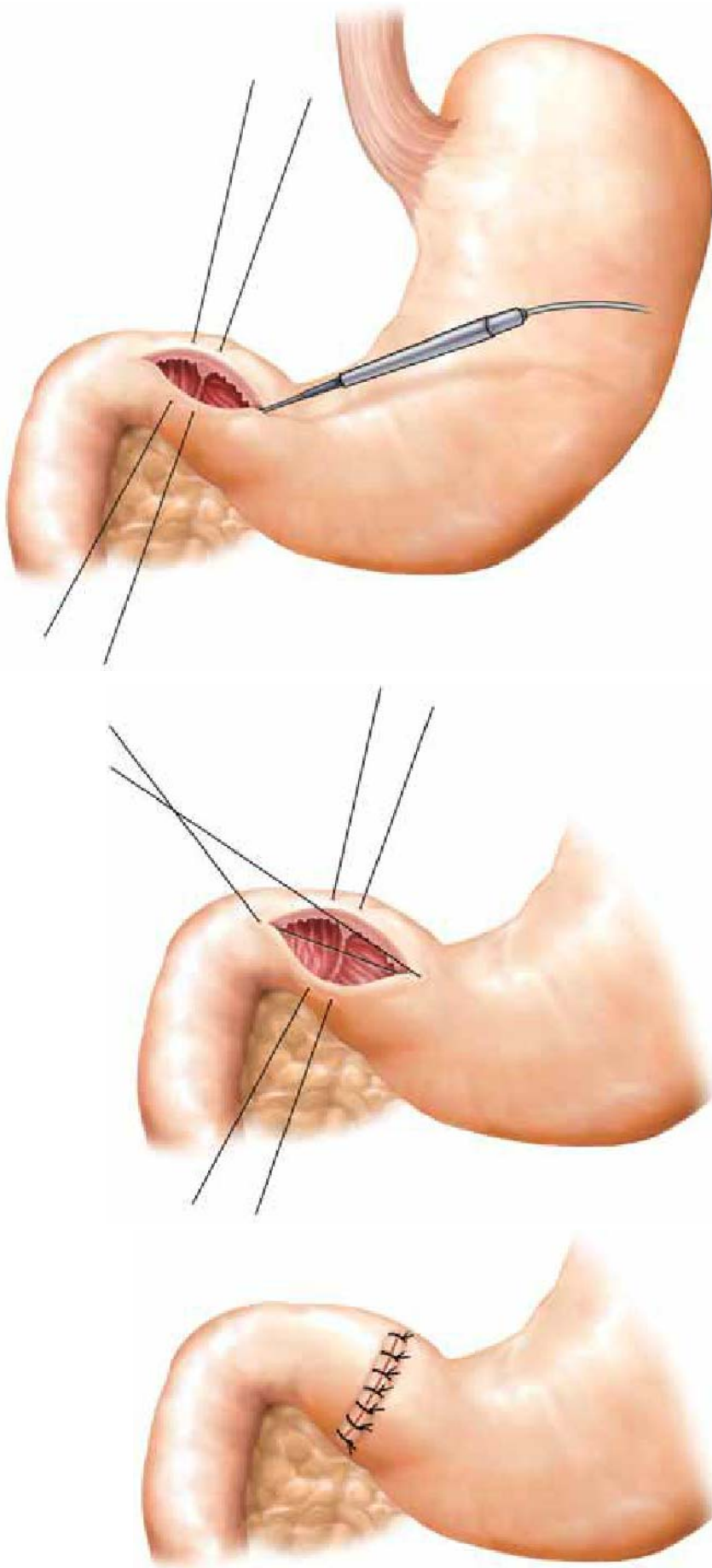


Figure 2.4 Schematic of Heineke–Mikulicz pyloroplasty. The pylorus is incised longitudinally and closed transversely.



Figure 2.5 Pylorus incised and opened. Stay sutures are placed at each corner to hold the pylorus open.

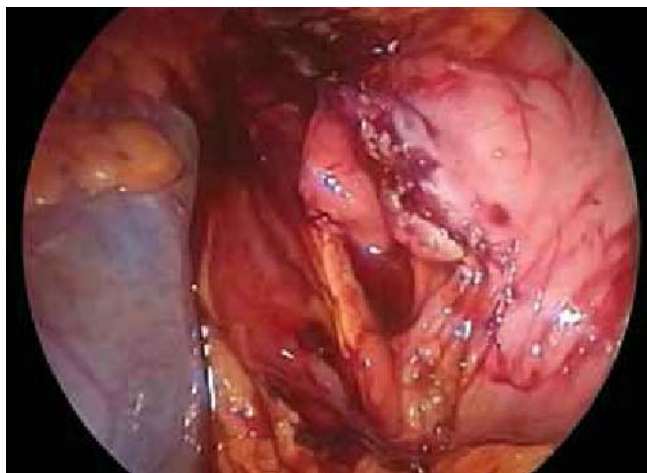


Figure 2.6 Pylorus pulled open prior to closure. The pylorus is incised completely from the prepyloric region into the duodenum.



Figure 2.7 Pylorus closed transversely and defunctionalized.

technique. This technique works well if the pylorus is short and soft without scarring. The enterotomy through the pylorus can be closed transversely with a linear stapler. The thickness and the ends of the staple line may need oversewing since the pylorus is of much thicker tissue than the other parts of the stomach and the duodenum. Integrity of the staple line may also be checked with blue dye.

Alternatively, a circular stapler can be used, where a portion of the pylorus is resected. This is usually a very quick procedure useful for patients with other medical comorbidities. An enterotomy is made in the body of the stomach anteriorly, large enough to fit a 25-mm circular stapler. The stapler is opened and positioned traversing the pylorus muscle with the distal end just in the first portion of the duodenum. The stapler is lifted anteriorly while simultaneously dunking the pylorus muscle posteriorly. Care is taken to avoid catching the pylorus posteriorly. The stapler is closed and fired, creating a partial resection of the pylorus muscle with opposition of the serosa. The staples through the pylorus are then reinforced with one suture on each end of the staple line. The stapler is then removed, and the patency of the pylorus is checked by directing the orogastric tube already in place or passing a flexible dilator through the gastric enterotomy distally into the duodenum. The enterotomy in the body of the stomach is closed using a linear stapler. Although one drawback in this technique is the creation of two separate staple lines, the staple line on the body of the stomach rarely causes a leak or stricture and is usually safe to perform. The integrity of both staple lines can be checked using blue dye as described above.

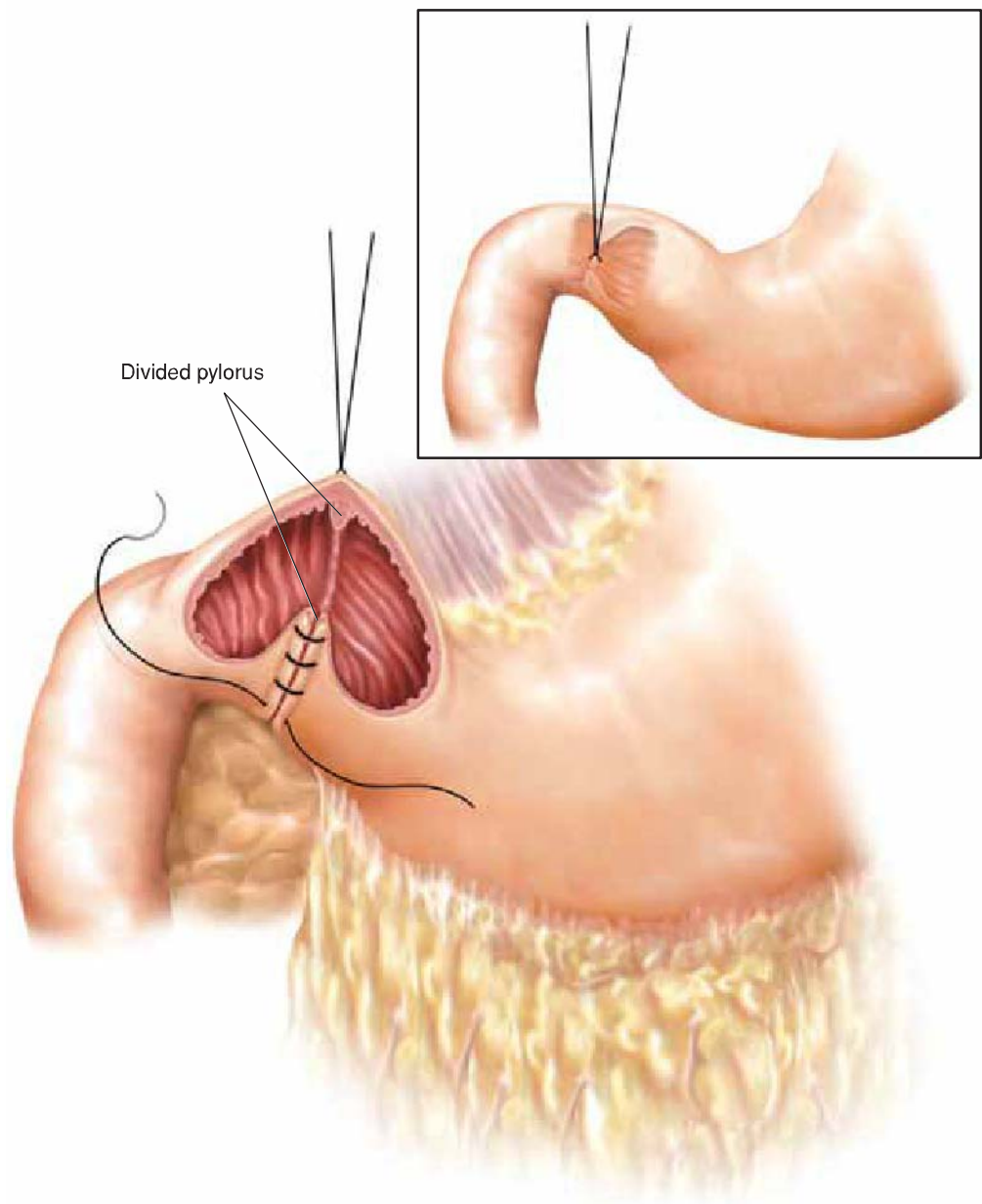
Although rarely necessary, Jaboulay pyloroplasty, a gastroduodenostomy in actuality, can be performed if significant scarring exists making HM pyloroplasty difficult with a high chance of leakage. Again, the first step is always a Kocher maneuver to mobilize the duodenum. The duodenum is apposed to the antrum along the greater curvature of the stomach with seromuscular bites of 3-0 silk suture, usually in an interrupted fashion. Matching enterotomies approximately 5 cm long are made in the stomach and duodenum just anterior to the apposed serosa. The end sutures are left long to hold as stays. The anastomosis is then created by suturing the duodenum to the stomach with full thickness bites of tissue to include mucosa. This is done in a running fashion along this posterior wall. A new stitch is run from the corner anteriorly half the length of the enterotomies, while another stitch is taken from the opposite corner. This anterior stitch can be an inverting stitch to oppose the serosa. If not, we suggest interrupted Lembert stitches anteriorly using 3-0 silk suture. Again a leak test is performed with blue dye in the stomach down the orogastric tube.

Finney pyloroplasty is rarely performed today but allows excellent drainage for chronically obstructed stomach. A Kocher maneuver is performed, followed by traction sutures of 3-0 silk as with HM pyloroplasty. A 10 to 12 cm inverted U-shaped incision is made from the lumen of the stomach, across the pylorus, and into the second portion of the duodenum (Fig. 2.8). A two-layer closure is performed. First, the posterior septum between the posterior walls of the stomach and duodenum are sutured, using a 3-0 continuous absorbable hemostatic suture (Fig. 2.9). This suture begins at the pyloric aspect of the posterior wall and continues anteriorly to secure the anterior walls of the stomach and duodenum (Fig. 2.10). An outer layer of 3-0 silk Lembert sutures completes the procedure (Fig. 2.11).

Alternatively, a stapled technique can be performed here as well. After mobilization of the duodenum, enterotomies are made in the duodenum and stomach approximately 5 cm distal and proximal to the pylorus, respectively. Limbs of the linear cutting stapler are inserted into the openings positioned toward pylorus, and the stapler is fired. The common enterotomy is then closed with linear stapler and excess tissue is trimmed and the staple line is checked for hemostasis and integrity. Again, a leak test can be performed.

Postoperative nasogastric decompression is used occasionally, especially if there is a gastric outlet obstruction from long-standing disease and scarring. Closed suction drainage is rarely used.

Figure 2.8 Recommended incision created for a Finney pyloroplasty, with superior and inferior stay sutures.



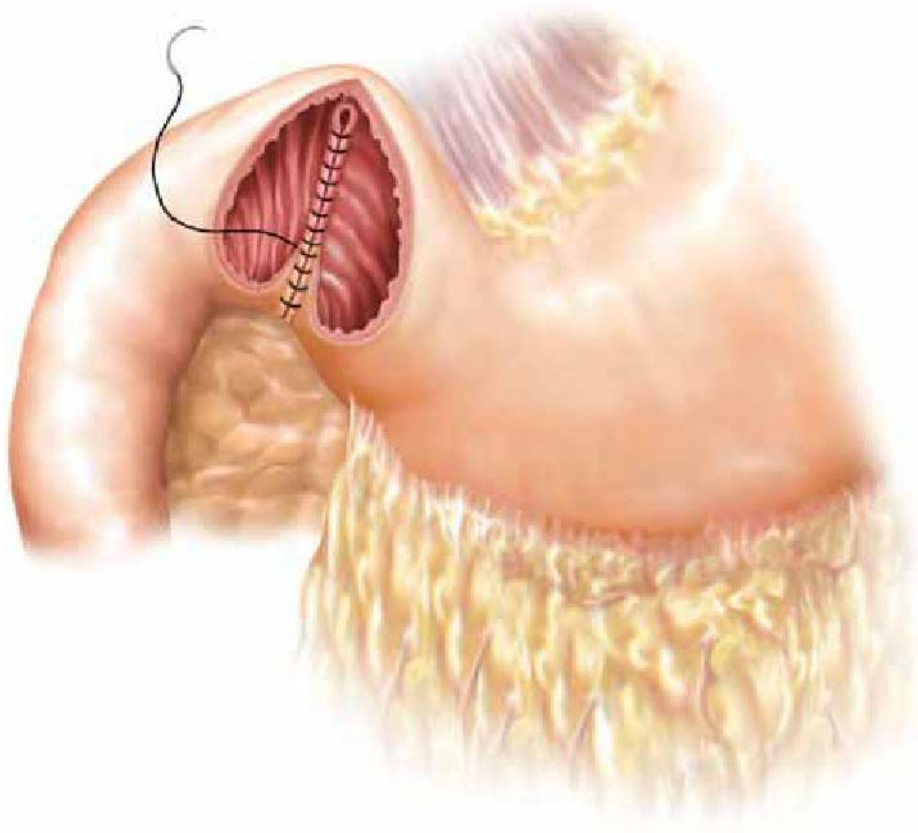


Figure 2.9 The inner row of running suture is usually of a 3-0 absorbable variety in the Finney pyloroplasty.

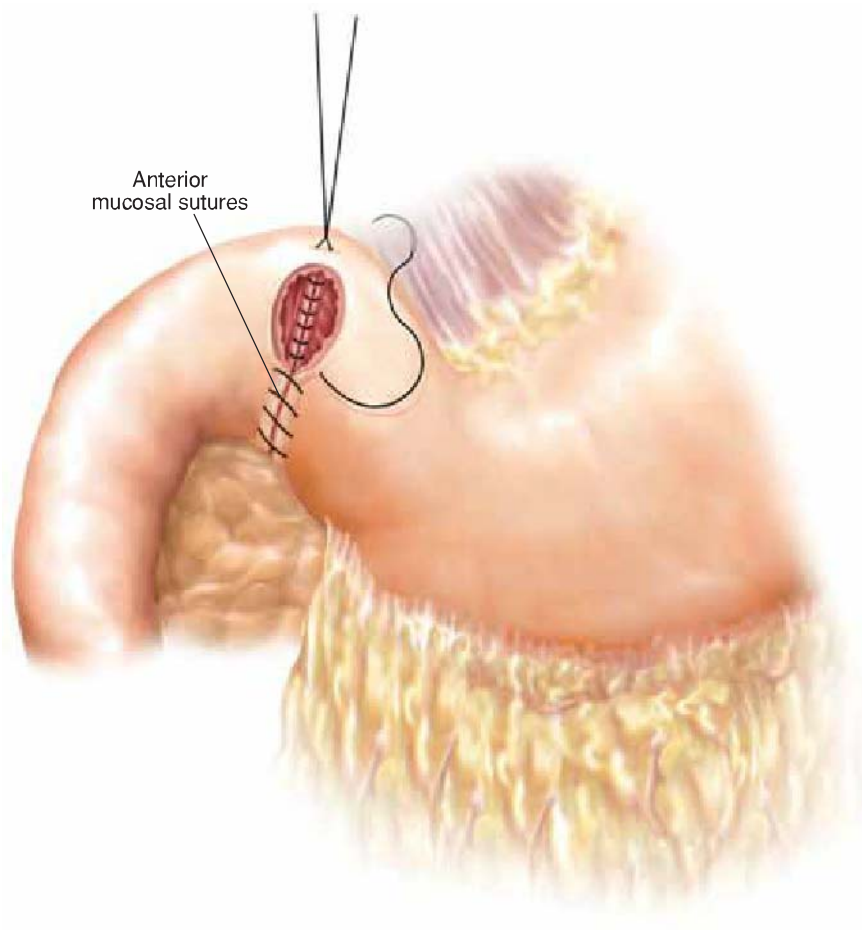


Figure 2.10 The running inner suture line continues from superiorly and inferiorly to end anteriorly.

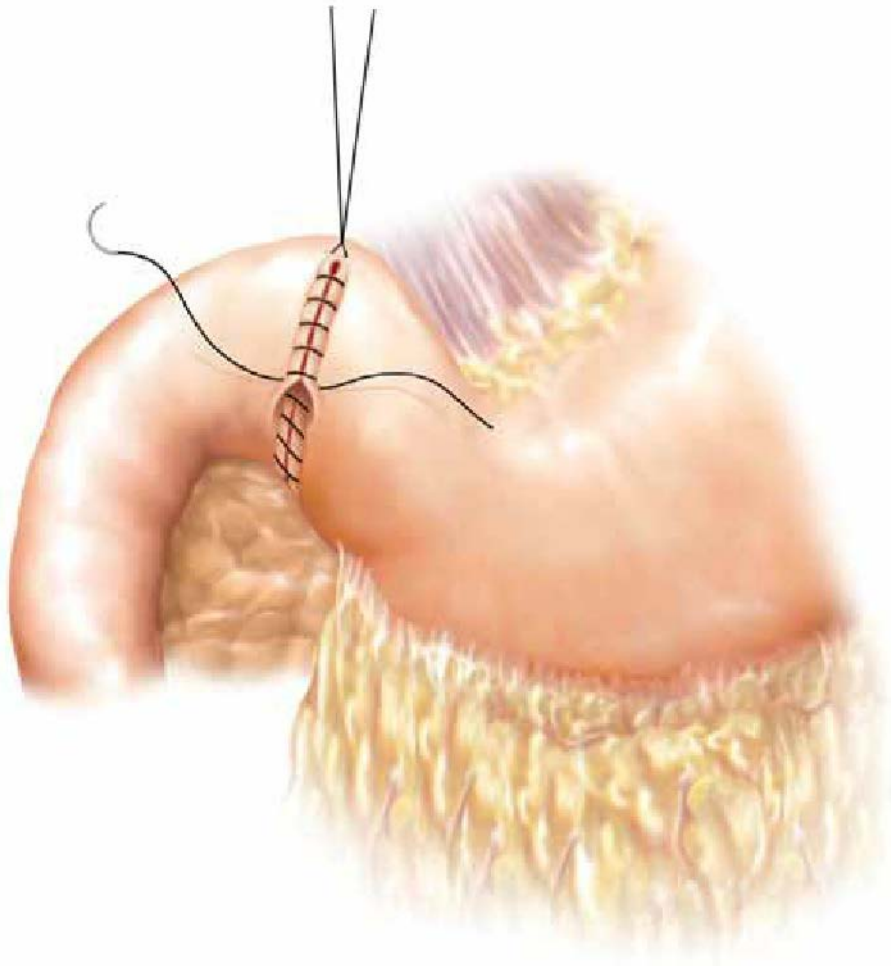


Figure 2.11 Interrupted 3-0 silk Lembert sutures completes the Finney pyloroplasty.



CONCLUSION

Although an uncommon procedure today, surgeons should have vagotomy and pyloroplasty in their operative armamentarium for treatment of peptic ulcer disease. Often, this procedure is necessary emergently, as in a bleeding ulcer refractory to medical management, and familiarity with this procedure will decrease operative time, which may be important for unstable patients.

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3 Truncal Vagotomy with Antrectomy and Billroth I Reconstruction

Stanley W. Ashley and Thomas E. Clancy



INDICATIONS/CONTRAINDICATIONS

Vagotomy has been recognized as a means of reducing gastric acidity, and thus it has been used in treating gastroduodenal ulcers since the early years of the 20th century. The use of vagotomy for peptic ulcer disease was found to be associated with delayed gastric emptying, so a gastric drainage procedure was added. The combination of truncal vagotomy and antrectomy, whereby the gastrin-producing antrum is removed, was subsequently proposed. Alternatives to truncal vagotomy to spare the celiac and hepatic branches (proximal selective vagotomy) or with preservation of vagal innervations of the antrum (highly selective vagotomy) have been proposed to mitigate postvagotomy complications of dumping and diarrhea. Still, the loss of receptive relaxation and gastric accommodation with all procedures makes some drainage procedure desirable.

The past few decades have seen a dramatic decrease in the role of surgical management of uncomplicated peptic ulcer disease. While operative management was previously routine for gastric and duodenal ulcers, the use of acid-suppressing medications such as histamine-2 receptor antagonists and proton pump inhibitors, as well as the treatment of *Helicobacter pylori* infection, has made many such procedures rare. Surgical management of peptic ulcer disease is now predominantly limited to complications of the disease, such as hemorrhage, perforation, and obstruction. This approach largely avoids the use of acid-reducing operations. Still, acid-reducing procedures may play an important role in the management of complicated and refractory ulcer disease, particularly in patients who have failed acid-suppressing medications.

Truncal vagotomy remains an important option for complicated peptic ulcer disease, particularly in the acute setting with a medication-refractory patient. Antrectomy will remove approximately 35% of the stomach, including all of the nonparietal cell portion of the stomach. This will allow resection of a distal gastric or prepyloric ulcer. Given very effective antisecretory medications and antibiotics to treat *H. pylori*, truncal vagotomy and antrectomy is primarily useful in the setting of pyloric outlet obstruction with recurrent ulcer symptoms or peptic ulcer disease complicated by bleeding or

perforation. Reconstruction of intestinal continuity can be performed via gastroduodenostomy (Billroth I) or loop gastrojejunostomy (Billroth II).

The indications for truncal vagotomy, antrectomy, and Billroth I gastroduodenostomy are as follows:

- prepyloric ulcer
- gastric ulcer
- peptic ulcer disease with gastric outlet obstruction
- recurrent ulcer after treatment with antisecretory medications
- emergency treatment of complicated ulcer disease, in a patient who is refractory to antisecretory medications



PREOPERATIVE PLANNING

Preoperative evaluation should include detailed imaging and laboratory data as well as endoscopic evaluation. In the case of complicated peptic ulcer disease, emergency surgery may not allow extensive preoperative workup. Preoperative laboratory indices should include hematocrit, bleeding parameters, basic chemistries, and tests of nutritional reserve given the risk of preoperative malnutrition and postoperative ileus complicating major gastrointestinal surgery.

Endoscopy is essential for the evaluation of the patient with bleeding secondary to peptic ulcer disease; endoscopic management is often sufficient to manage bleeding without surgery. Surgery is required for bleeding in the unstable patient, after extensive transfusion (over 6 units of blood), or for rebleeding after initial endoscopic management. Precise localization of the bleeding source is important.

Endoscopy is also important in the diagnosis of *H. pylori* infection. In patients with refractory peptic ulcer disease, endoscopy is important for biopsy to rule out occult gastric malignancy. Workup should include biopsies of the ulcer base and surrounding gastric mucosa. Endoscopy also has a role in the documentation of healing of gastroduodenal ulcers after the initiation of antisecretory medications, as failure of ulcers to heal may portend occult malignancy.

Imaging to include a chest x-ray and CT scan of the abdomen is useful to detect potential metastatic disease when malignancy is considered.



SURGICAL TECHNIQUE

Pertinent Anatomy

The vagal nerves, a plexus around the intraabdominal esophagus, will join to form two trunks at the esophageal hiatus. The anterior (left) vagus nerve is positioned along the anterior wall of the esophagus, while the posterior vagus is found between the posteromedial wall of the esophagus and the right crus of the diaphragm (Fig. 3.1).

Positioning

Patients are positioned in the supine position. As access to the upper stomach and distal esophagus is critical, an upper midline incision to the xiphoid process is typically utilized. A bilateral subcostal incision, providing excellent exposure to the duodenum, may compromise optimal exposure of the upper abdomen. The abdomen is prepped from the low chest to the pubis.

Mild reverse Trendelenburg position is useful, and a nasogastric tube not only decompresses the stomach but will also allow easier identification of the esophagus. Mobilization of the left lobe of the liver by dividing the triangular ligament is performed selectively; although this maneuver may aid exposure of the gastroesophageal junction

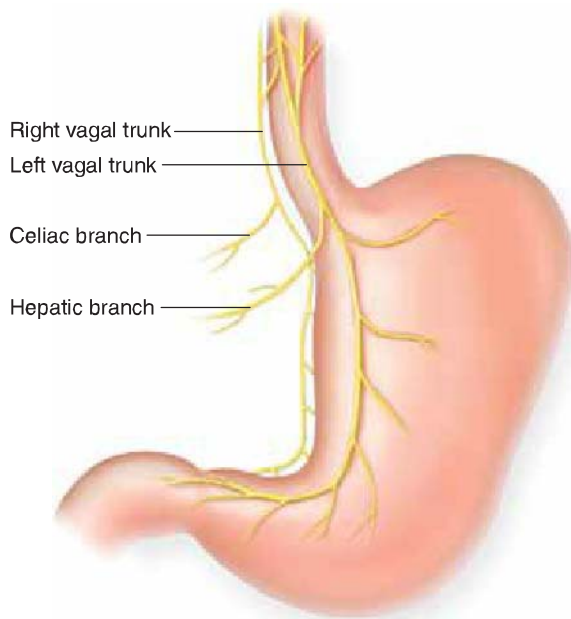


Figure 3.1 The vagal nerves from the intrathoracic esophagus form two major trunks before entering the abdomen at the esophageal hiatus.

in some patients, a large or redundant left lateral segment may be held in place by the triangular ligament and exposure may be impeded by its division in some patients.

Technique

Vagotomy

Mobilization of the liver medially or superiorly may be necessary for optimal visualization of the gastroesophageal junction. The left triangular ligament of the liver can be divided with cautery to facilitate medial rotation of the left lateral segment of the liver.

The distal esophagus is exposed by incising the peritoneal covering of the gastroesophageal junction with cautery, incising the peritoneum from the lesser curvature to the cardiac notch at the greater curvature. Gentle blunt dissection is used to surround the esophagus. A Penrose drain is placed around the distal esophagus for retraction. Care should be taken to place the Penrose drain at a sufficient distance from the esophagus to include the vagal trunks. The posterior vagal trunk may be palpated during this maneuver as a tight cord. For complete vagotomy, the distal esophagus must be mobilized proximally and stripped of peritoneal attachments for approximately 5 cm.

The operator should note the clockwise rotation of vagal trunks as the vagi pass from the thoracic cavity into the abdominal cavity. Exposure of the anterior vagal trunk is facilitated by inferior–posterior traction on the Penrose drain. Similarly, the posterior vagus may run along the posterior-right side of the distal esophagus, and exposure is facilitated by retracting the Penrose drain into the wound and toward the left of the incision. A single anterior vagal trunk is usually identified in the anterior midportion of the esophagus, 2 to 4 cm above the gastroesophageal junction. It is not uncommon for vagal fibers to be distributed among two or three smaller cords at this level. These trunks are individually lifted up and 2- to 4-cm segments of each are separated from the surrounding tissues. The nerves are surrounded, clipped on two sides, and a 2-cm middle portion is excised and sent to pathology to confirm the presence of the nerve via frozen section. The posterior vagal trunk is usually identified along the right edge of the esophagus. If the anterior vagus has already been divided, the esophagus is more mobile. This mobility allows downward traction on the gastroesophageal junction, which allows the posterior vagus to “bowstring” and makes it easier to identify. A 2- to 4-cm segment is separated from the surrounding tissues, its margins marked with clips, resected, and sent to pathology for frozen section (Figs. 3.2 and 3.3).

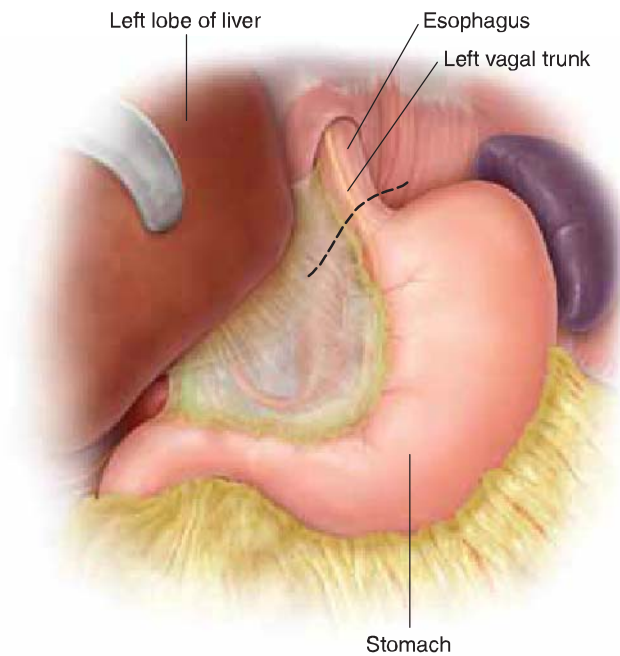


Figure 3.2 The peritoneum over the gastroesophageal junction is divided with cautery to expose the esophagus. The left lateral segment of the liver may require superior or medial retraction for exposure.

Antrectomy

After vagotomy is performed, antrectomy is a useful drainage procedure, particularly in the setting of duodenal or pyloric channel ulcer with gastric outlet obstruction. Antrectomy removes approximately 35% of the stomach to include the gastrin-producing mucosa. Useful landmarks include the incisura angularis on the lesser curvature and a point between the pylorus and the fundus on the greater curvature where the right and the left gastroepiploic vessels meet (watershed area).

The lesser sac is entered via the gastrocolic ligament, an avascular plane in the omentum. Dissection should begin at the mid-greater curvature and continue along the greater curvature of the stomach toward the pylorus. Adhesions between the posterior wall of the stomach and the pancreas are divided sharply. Near the pylorus, the right gastroepiploic vessels are divided and ligated. Dissection should be carried about 1 cm

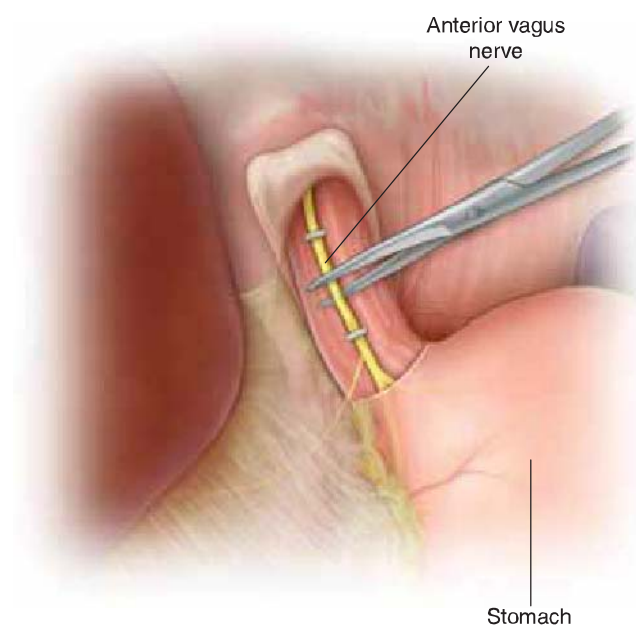


Figure 3.3 Gentle traction to the stomach facilitates identification of the vagus nerves. Clips are applied to each trunk proximally and distally, with removal of a 2- to 3-cm segment between the clips.

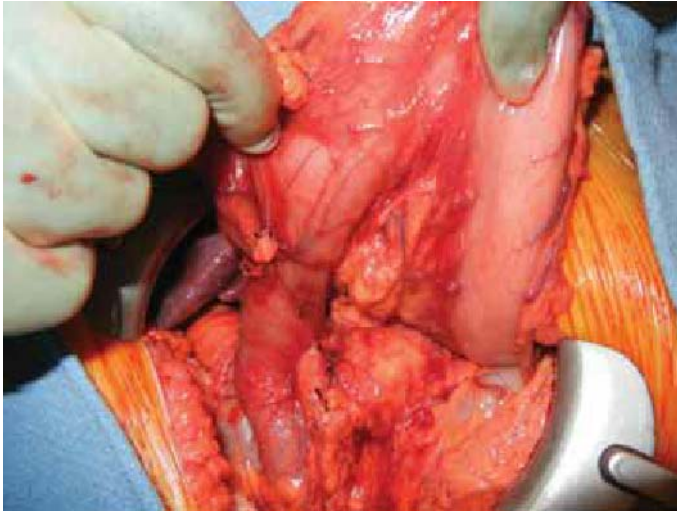


Figure 3.4 The stomach has been mobilized, with division of the gastrocolic ligament and right gastroepiploic vessels. The pancreas is visible at the base of the lesser sac.

past the pylorus, taking care to avoid injury to the pancreas. Along the lesser curvature between the incisura and the pylorus, tissues including distal branches of the left gastric artery are divided and ligated. The right gastric artery is similarly divided and ligated.

If a Billroth I anastomosis is planned, it is essential to mobilize the duodenum thoroughly with a generous Kocher maneuver. The peritoneum along the lateral duodenum should be divided with cautery. Dissection should begin near the porta hepatis, taking care to avoid injury to the common bile duct and portal vein. The Kocher maneuver should include the peritoneum along the inferior border of the third portion of the duodenum, separating the third portion of the duodenum from the transverse mesocolon to allow mobilization of the duodenum superiorly as well as laterally. If duodenal scarring prevents Billroth I anastomosis, a Kocher maneuver is not required.

A GIA linear stapler is used to divide the stomach at the estimated border of the antrum. The antrum is retracted anteriorly and to the patient's left side. The duodenum is divided with a TA stapler. Particular care must be taken with a thickened and inflamed duodenum in the setting of chronic scarring or perforation. For thick tissue, primary suture closure is preferable to stapled closure, and omental patch reinforcement of the duodenal stump may be helpful (Figs. 3.4–3.7).

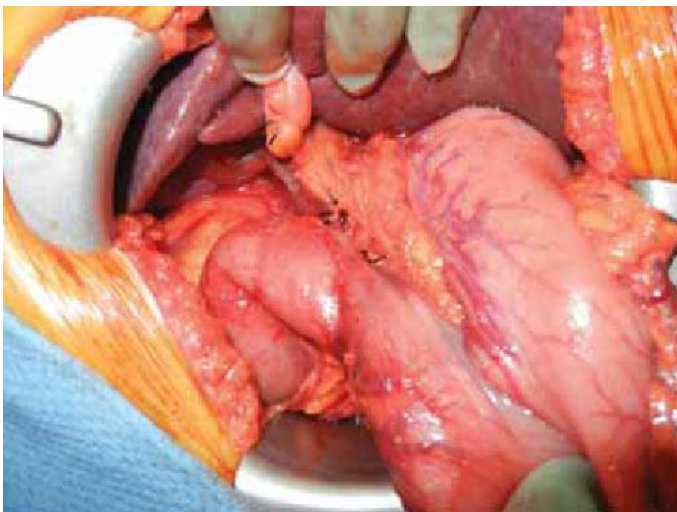


Figure 3.5 A full Kocher maneuver has been performed to mobilize the duodenum medially and superiorly.

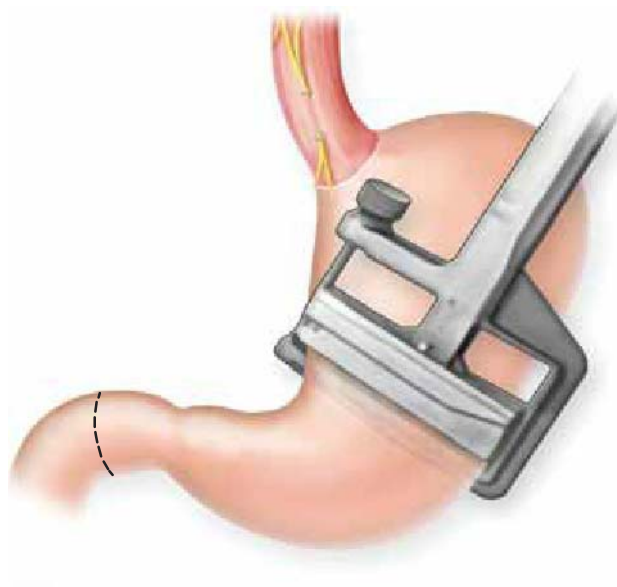


Figure 3.6 After entering the lesser sac, the stomach is divided proximally between the incisura and midway along the greater curvature to include the entire antrum.

Billroth I Gastroduodenostomy

Primary gastroduodenostomy, or Billroth I reconstruction, is preferable to allow preservation of antegrade flow. The gastroduodenostomy is created in two layers. A posterior row of interrupted seromuscular silk sutures are placed first in the Lembert manner between the duodenum and the stomach at the inferior/lateral border of the gastric staple line. The superior/medial portion of the duodenal staple line is removed with cautery, and the gastric staple line is opened a corresponding length. An inner layer of running 3-0 vicryl suture (or other absorbable suture) is placed, followed by an anterior outer row of interrupted 3-0 silk Lembert sutures. The junction of the anastomosis and the gastric staple line has been referred to as the “angle of sorrow” due to the complication of leakage at this intersection of suture/staple lines. This angle should be reinforced with additional 3-0 silk Lembert sutures.

Stapling techniques have been described for the gastroduodenostomy as well; the anvil of a 25-mm EEA stapler may be sewn into the duodenal stump with a purse-string suture. The circular stapler may be inserted through anterior gastrotomy, with the gastrotomy closed with a TA stapler (Figs. 3.8–3.10).



Figure 3.7 The stomach has been divided just proximal to the antrum, from the incisura to the mid-greater curvature. The duodenum is divided with a stapler 1 cm distal to the pylorus.

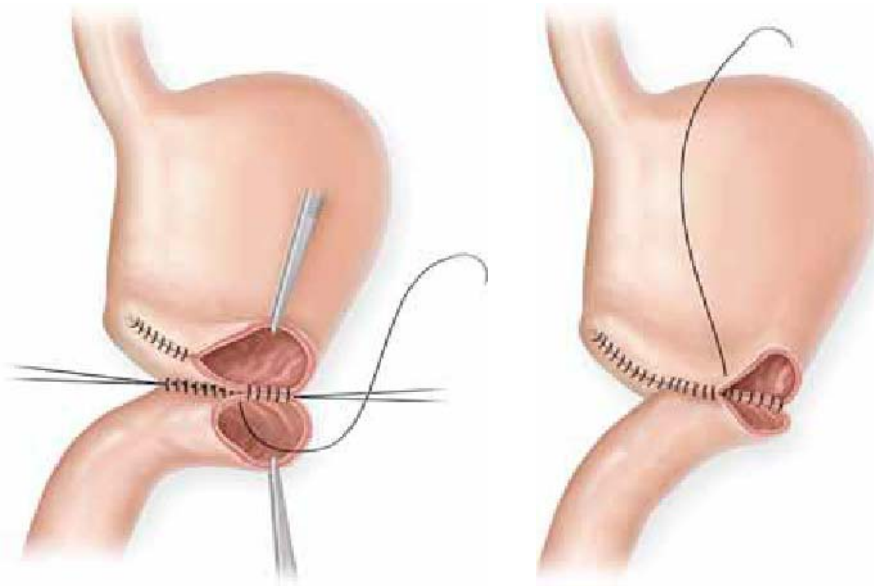


Figure 3.8 The gastrodudenostomy is created in two layers. The posterior layer is completed with 3-0 silk. An inner layer of running absorbable suture is placed, followed by an anterior row of interrupted 3-0 silk seromuscular sutures.

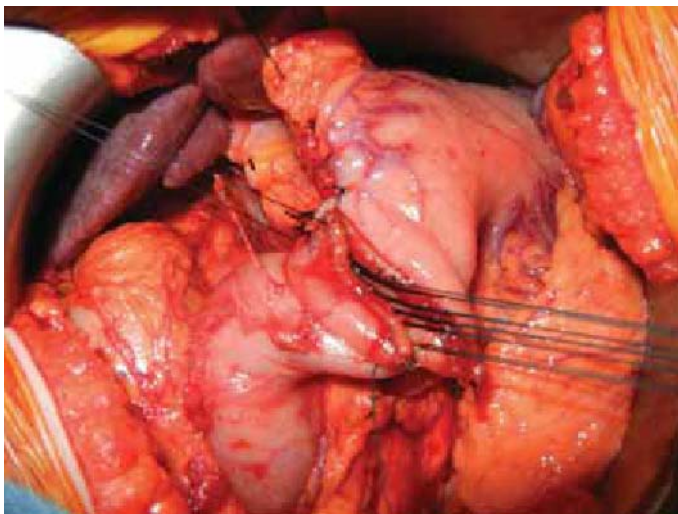


Figure 3.9 The Billroth I gastrodudenostomy is constructed first with a posterior row of interrupted seromuscular silk sutures. The gastric and duodenal staple lines have not yet been removed. The medial portion of the gastric staple line has been oversewn with seromuscular silk sutures.

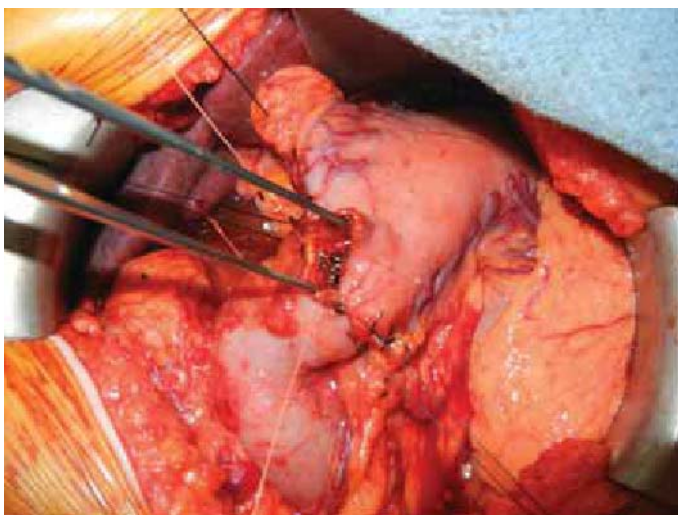


Figure 3.10 The Billroth I anastomosis is continued by removing the staple lines of the stomach and duodenum with cautery. An inner running layer of full-thickness 3-0 Vicryl suture is used in this case.



POSTOPERATIVE MANAGEMENT

Nasogastric suction is continued postoperatively until the volume of gastric effluent falls to less than 1,000 mL/24 hours and there is no clinical evidence of anastomotic leak. Typically patients will require nasogastric decompression for several days. Routine radiographic analysis via barium swallow is not necessary unless there is clinical suspicion for an anastomotic leak (Fig. 3.11).



COMPLICATIONS

An unfortunate complication after vagotomy is recurrent ulceration due to persistent vagal innervations. While some recurrences can be managed medically, some will require reoperation, particularly in an acute presentation such as perforation. Dense adhesions to the left lobe of the liver can make reoperative vagotomy difficult, and the best approach for repeat truncal vagotomy is via transthoracic or thoroscopic access.

Given the loss of vagally mediated receptive relaxation of the stomach, emptying of liquids is accelerated after vagotomy. Dumping, manifest as postprandial discomfort and dizziness, can be seen in 10% to 15% of patients after truncal vagotomy and antrectomy and can be disabling in 1% to 2% of patients. “Early” dumping may occur soon after a meal and include discomfort, dizziness, syncope, and palpitations. “Late” dumping occurs several hours postprandially and may include hypoglycemia. Dietary changes are effective in mitigating symptoms for most patients. Somatostatin analogues are also reported to have a beneficial effect on symptoms of dumping when administered prior to meals. Delayed gastric emptying after gastrojejunostomy is managed conservatively. Refractory delayed gastric emptying after Billroth I reconstruction is managed with gastrojejunostomy to the greater curvature of the stomach to avoid devascularization and duodenal stump leakage. Bleeding after partial gastrectomy is generally seen as intraluminal blood loss. An endoscopic approach with coagulation or epinephrine injection of the anastomosis is preferable to surgical exploration.

Alkaline reflux gastritis is common after partial gastrectomy, affecting 5% to 15% of patients. Reflux is common, although pain and nausea are less commonly seen. Endoscopy is important in the setting of postoperative epigastric discomfort to rule out recurrent ulceration. Endoscopy may show bile reflux into the stomach, as well as gastritis. HIDA scan (99mTc hepatic iminodiacetic acid scan) can also be used to establish the diagnosis. Medical management, including antacids, bile acid chelators, and dietary

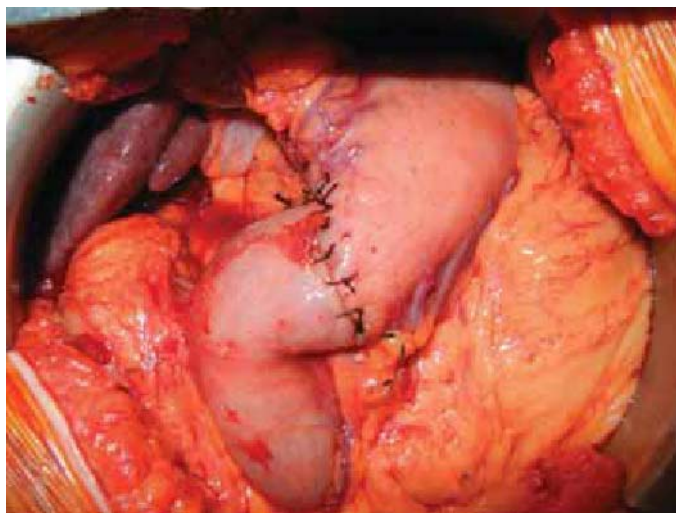


Figure 3.11 The Billroth I anastomosis is completed with an anterior row interrupted seromuscular 3-0 silk sutures.